Cloth Self Collision with Predictive Contacts

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EA Frostbite

No Mathematics Contained herein!

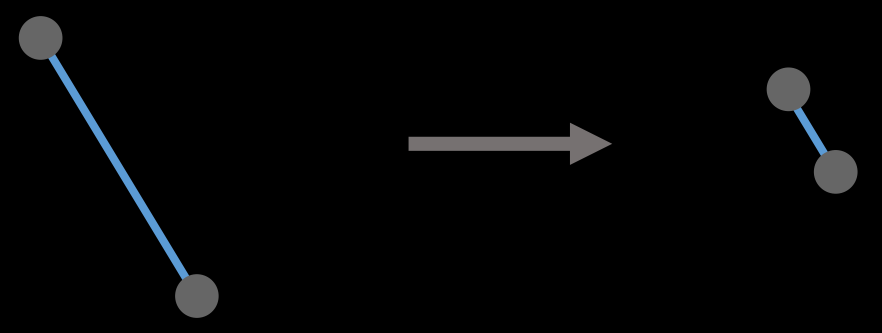
Get the technical report:
https://www.ea.com/frostbite/news/cloth-self-collision-with-predictive-contacts

Self Collision: Huh?




Simulating Cloth: How

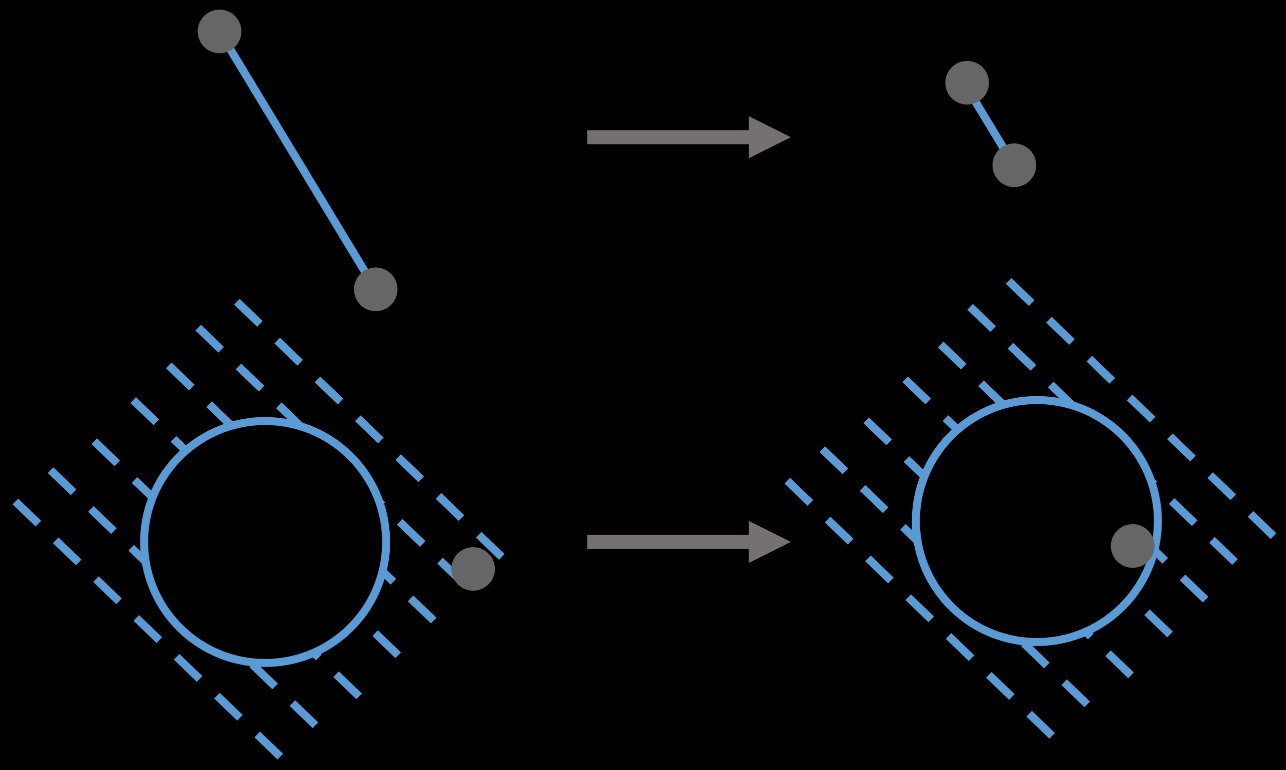
Length Constraints



Simulating Cloth: How

Length Constraints

Anchor Constraints

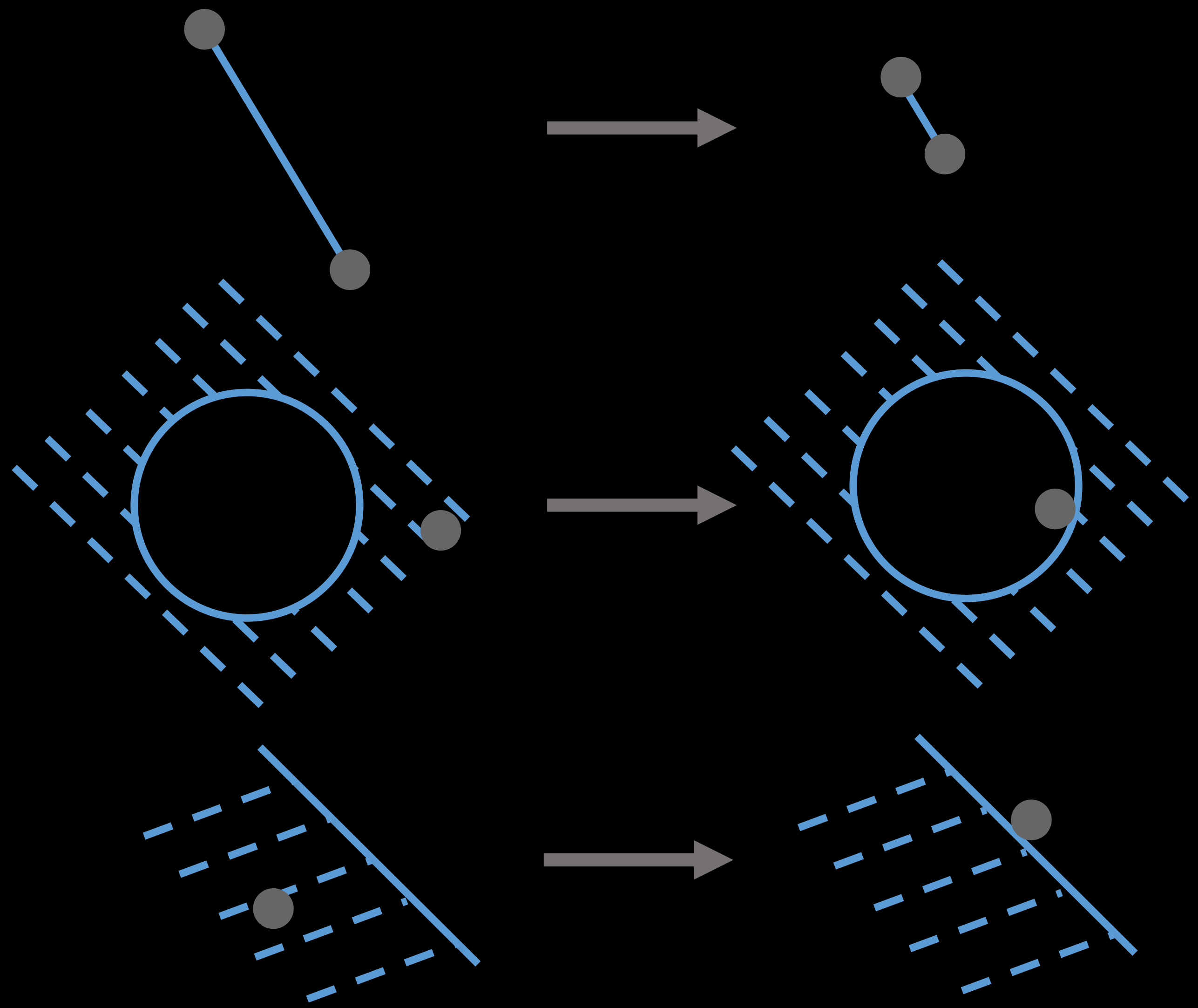


Simulating Cloth: How

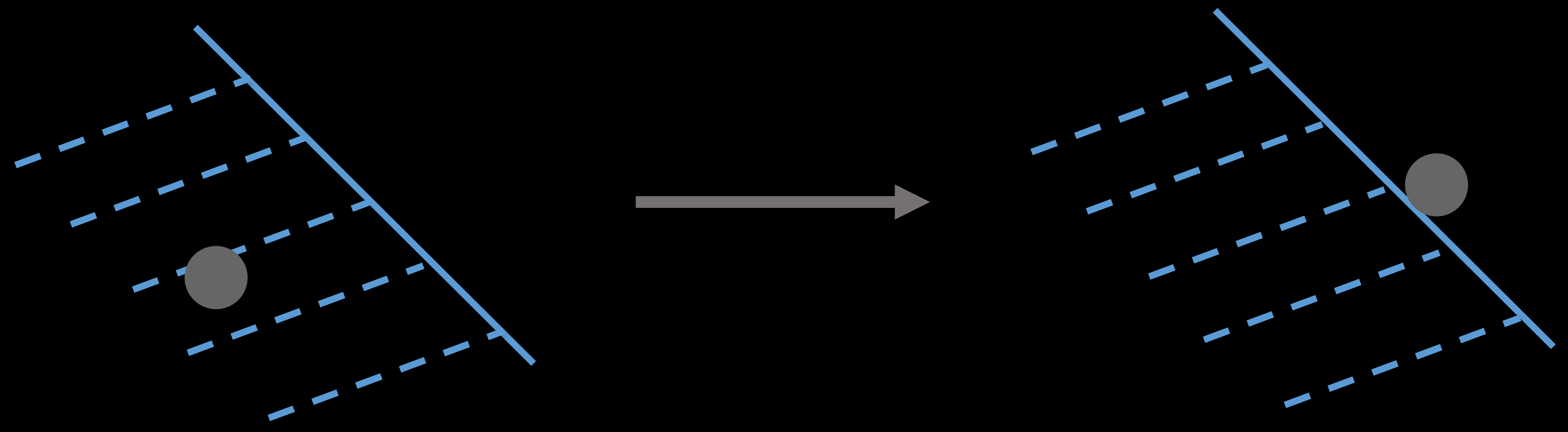
Length Constraints

Anchor Constraints

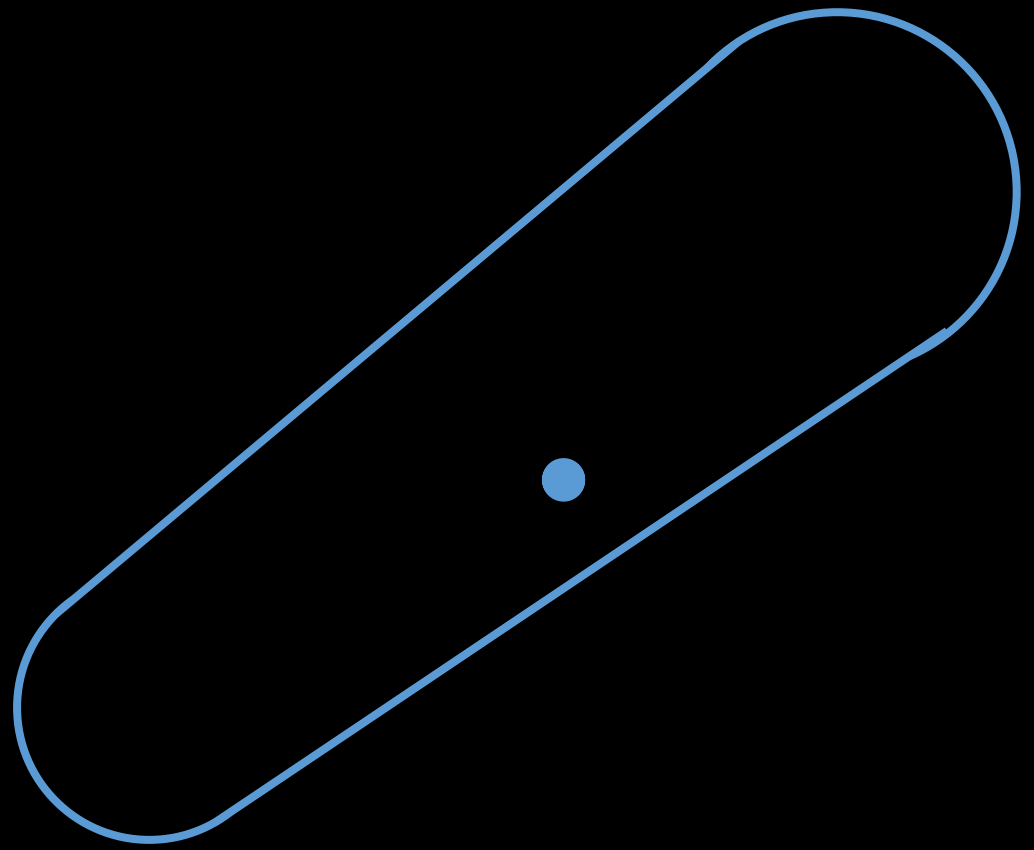
Contact Constraints



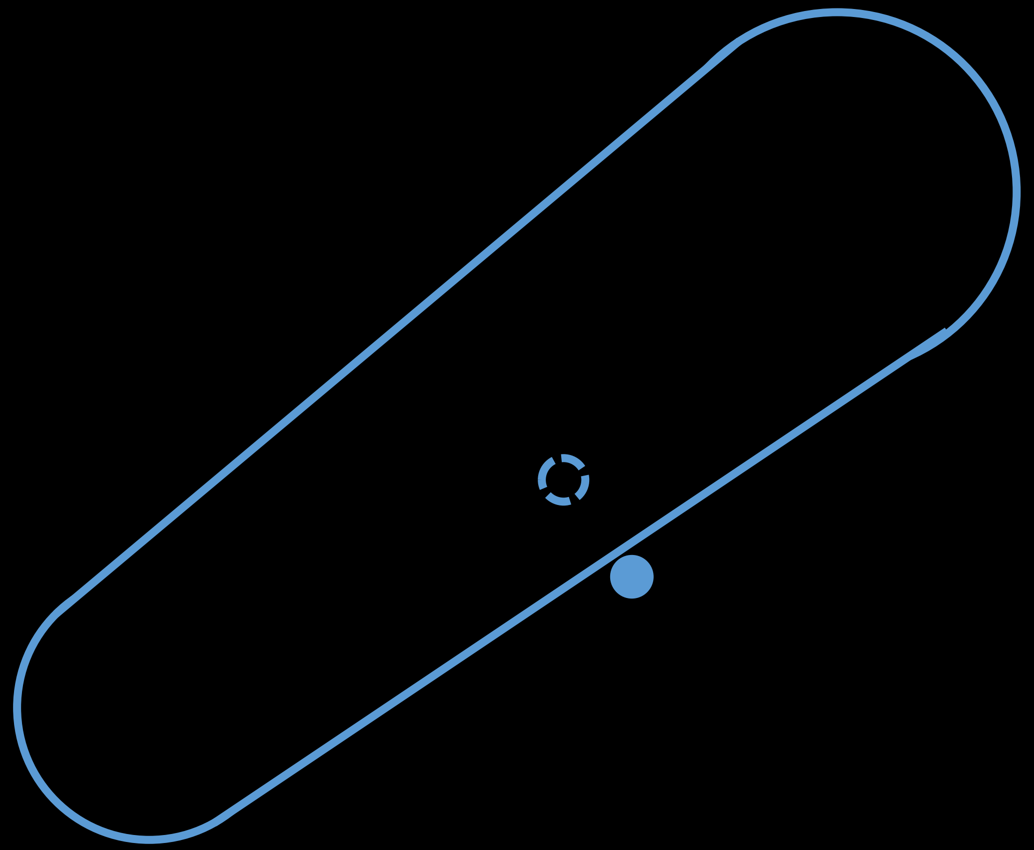
Collision



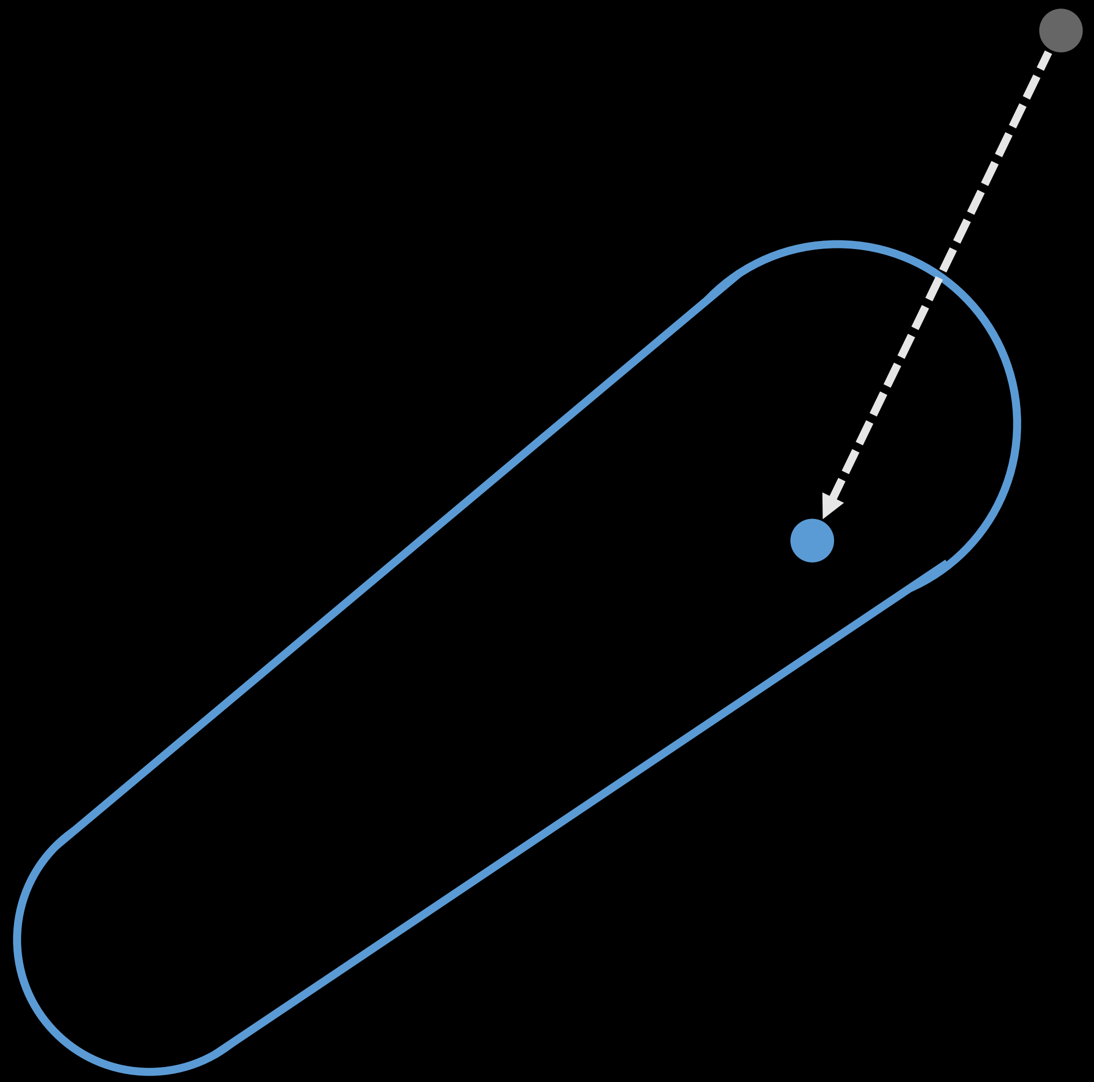
Discrete Collision



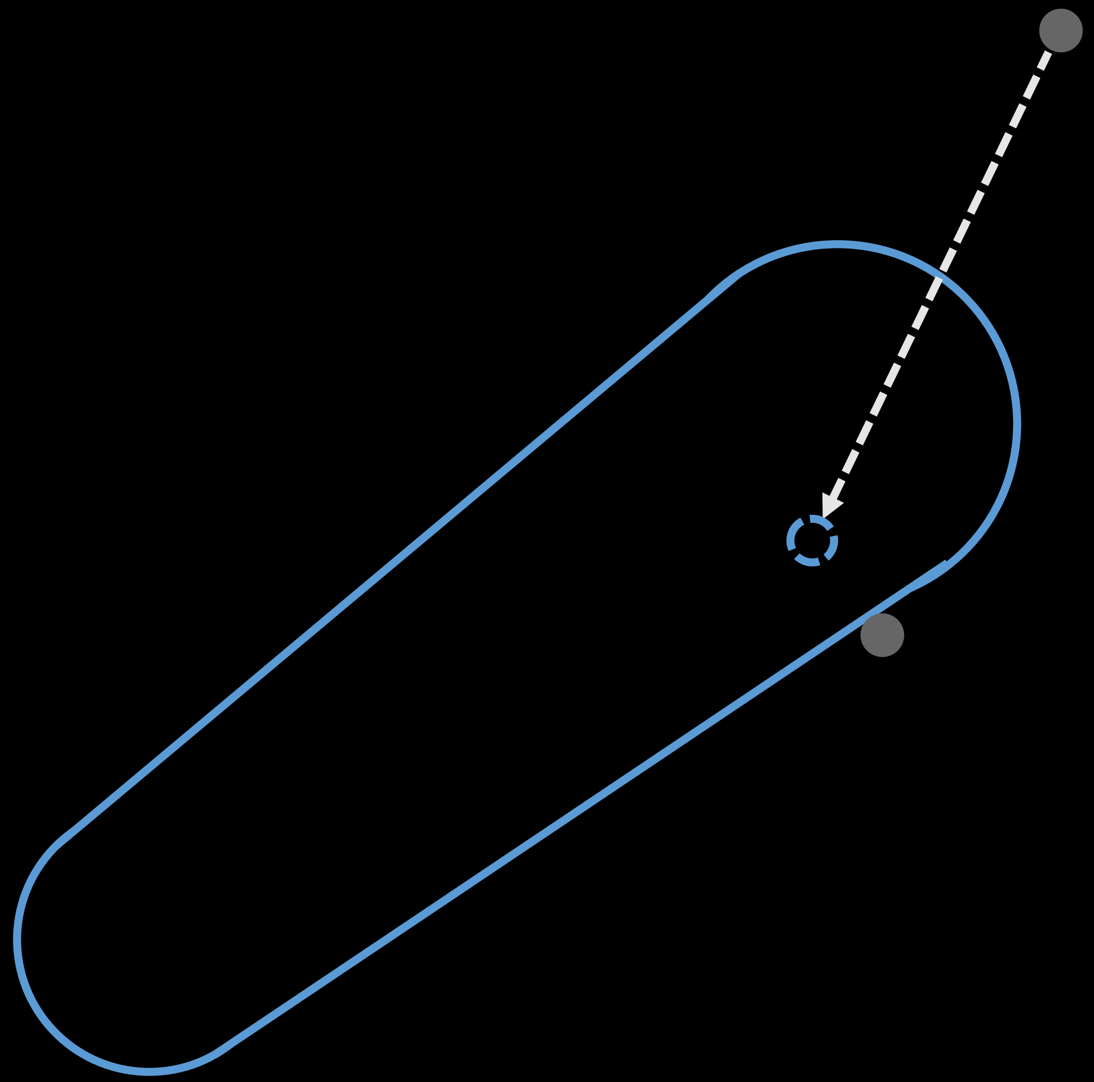
Discrete Collision



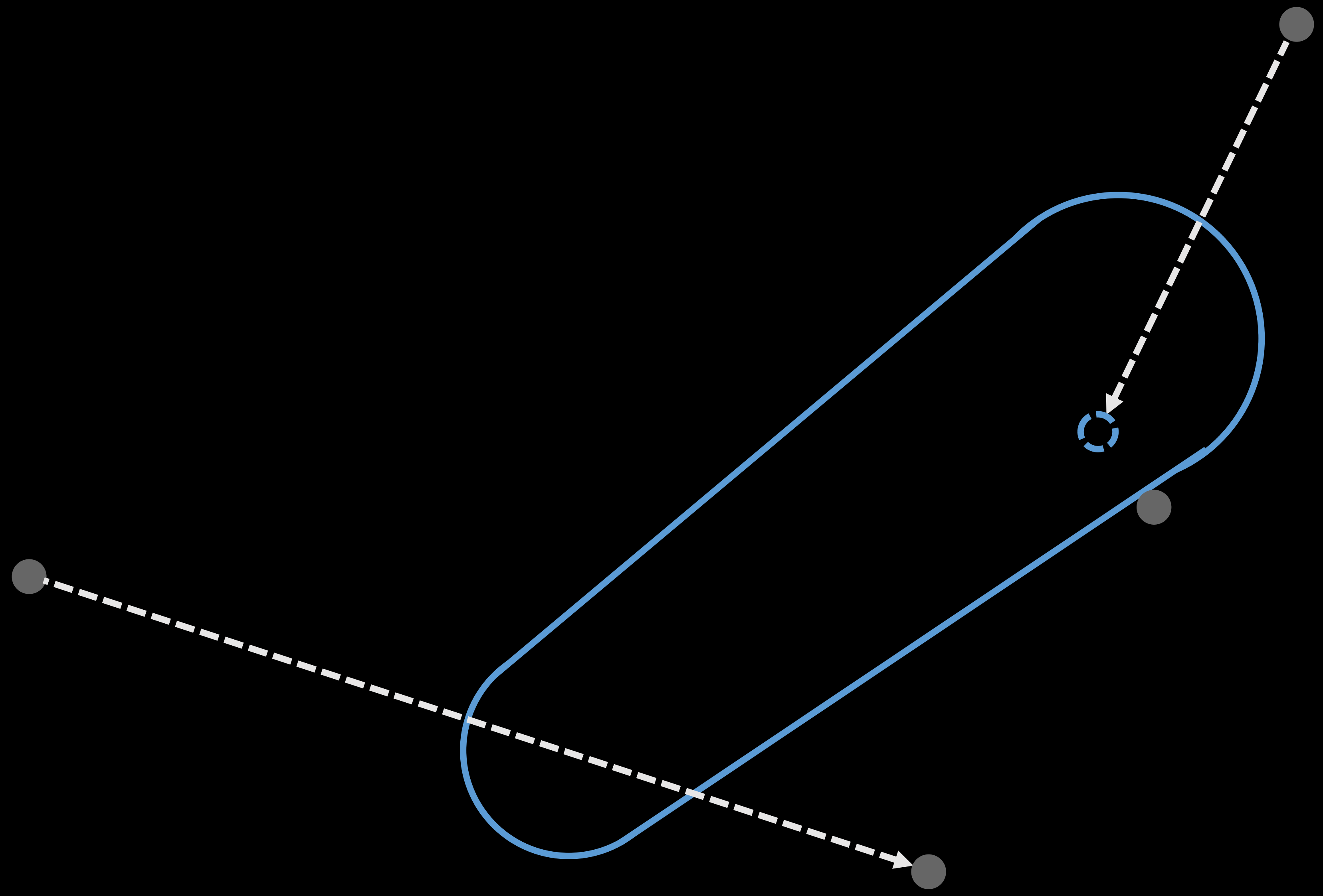
Discrete Collision



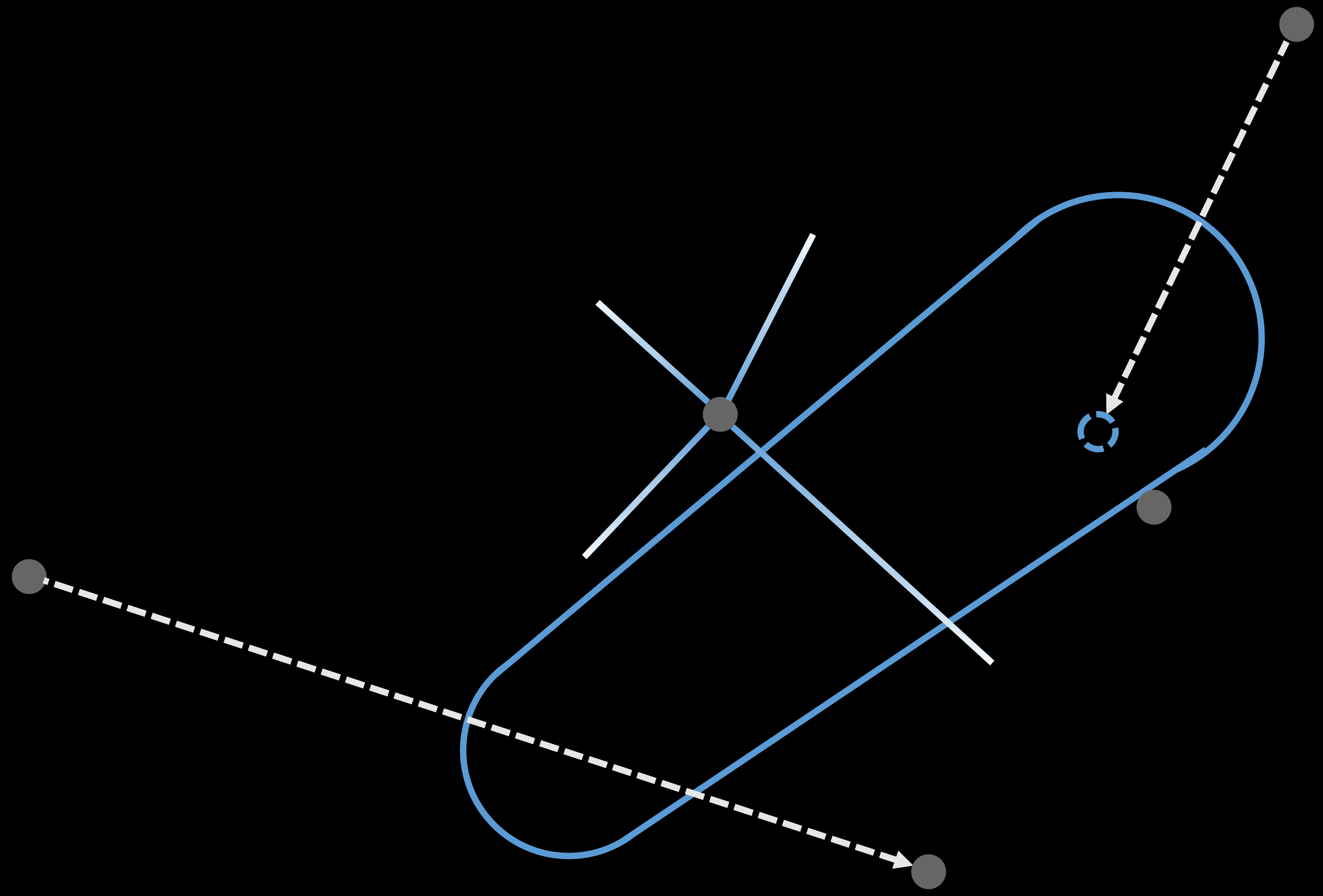
Discrete Collision



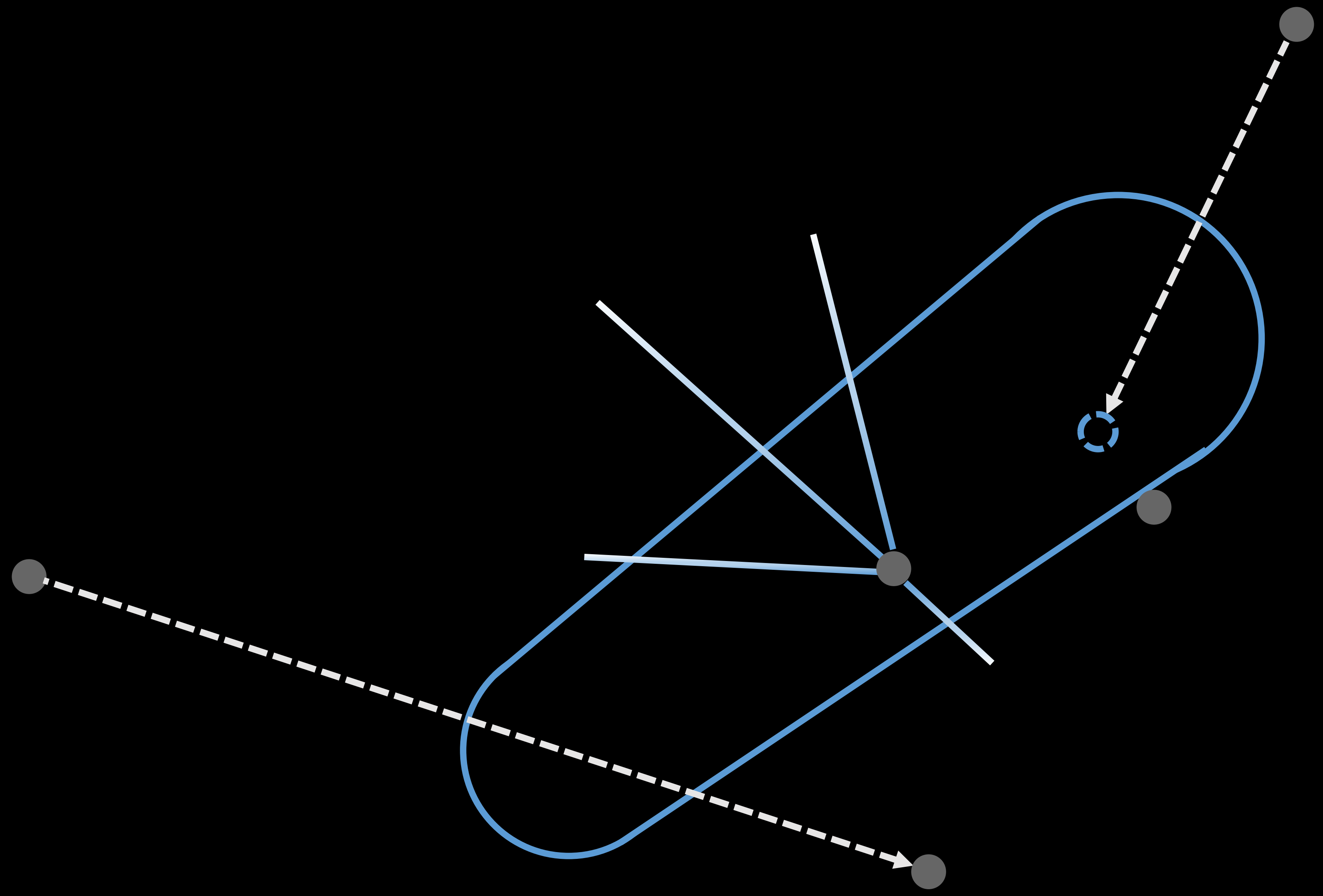
Discrete Collision



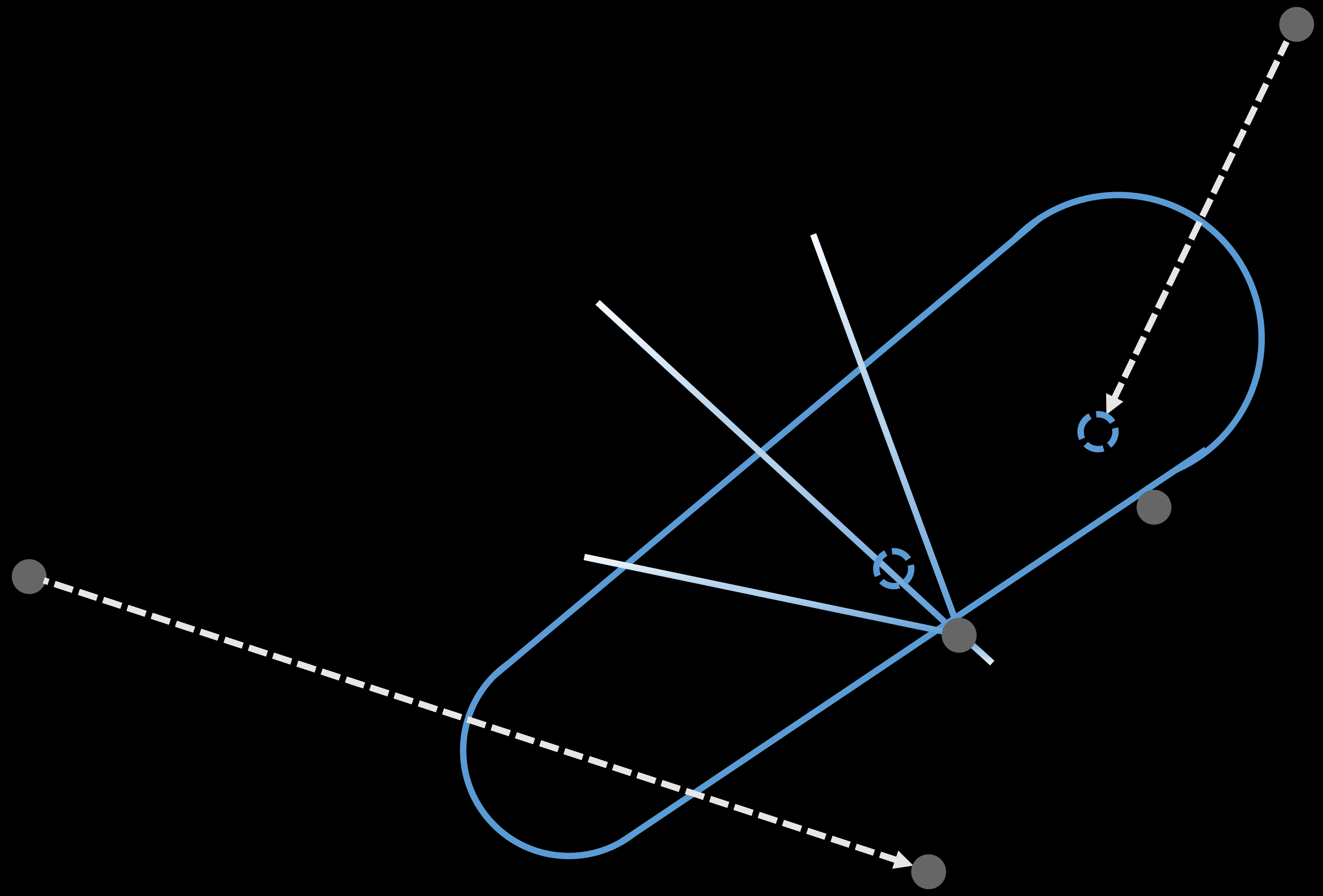
Discrete Collision



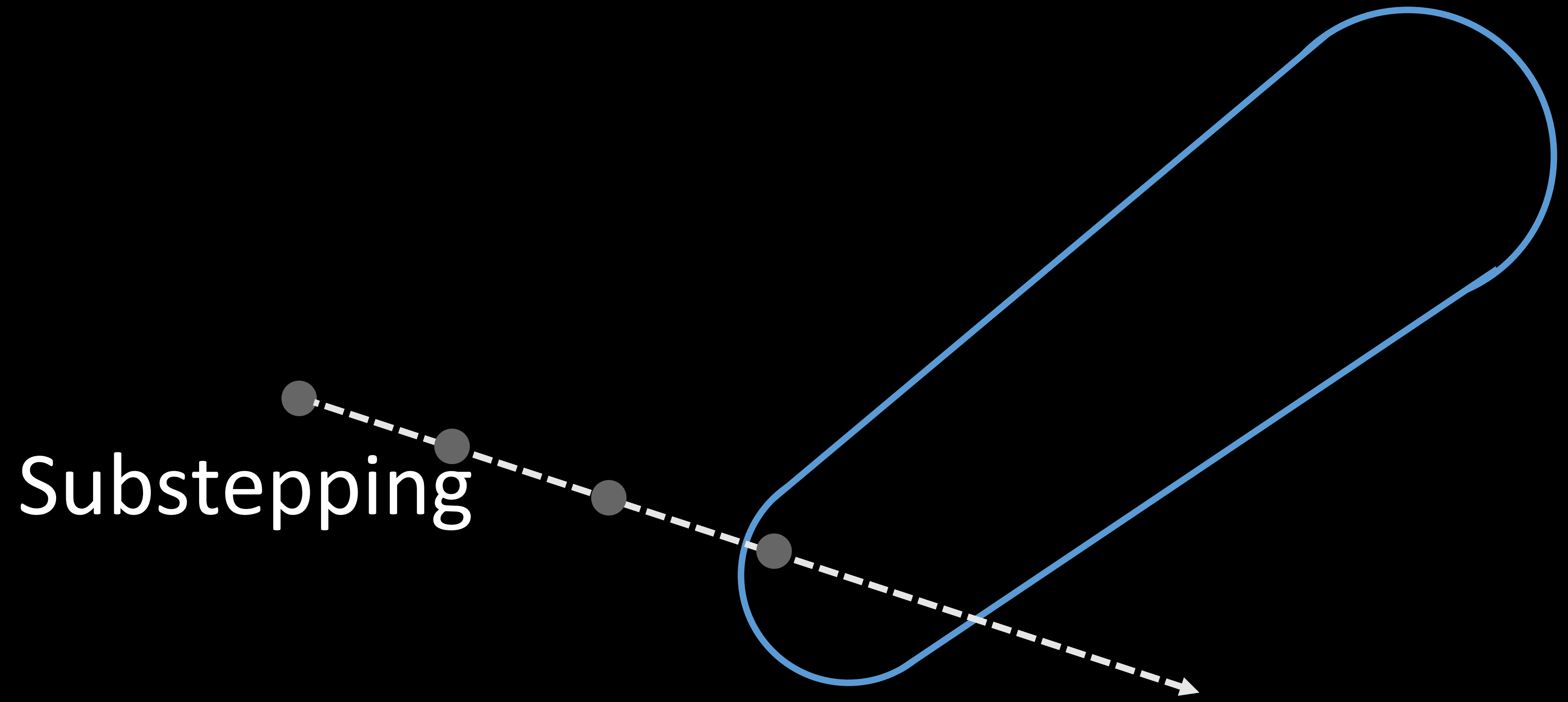
Discrete Collision



Discrete Collision



Continuous Collision



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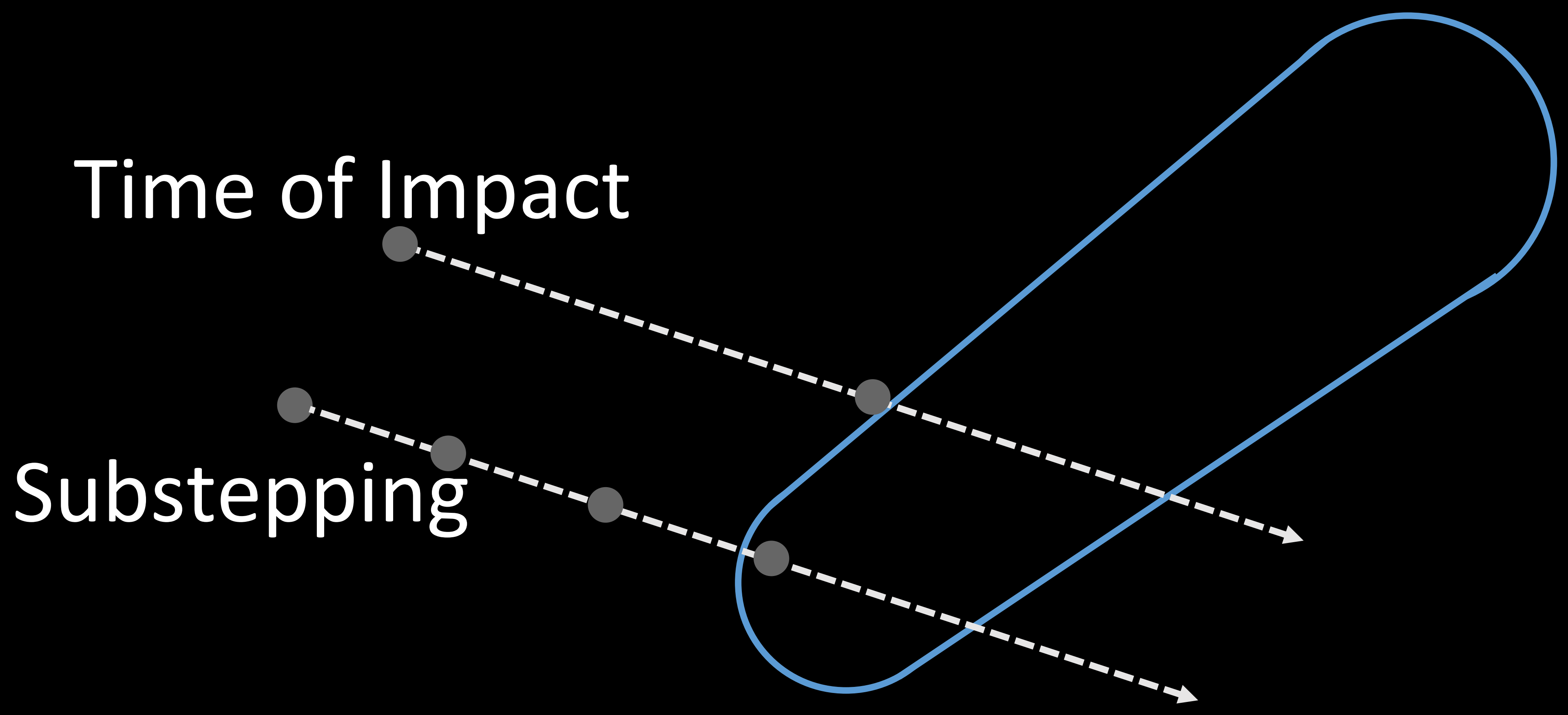

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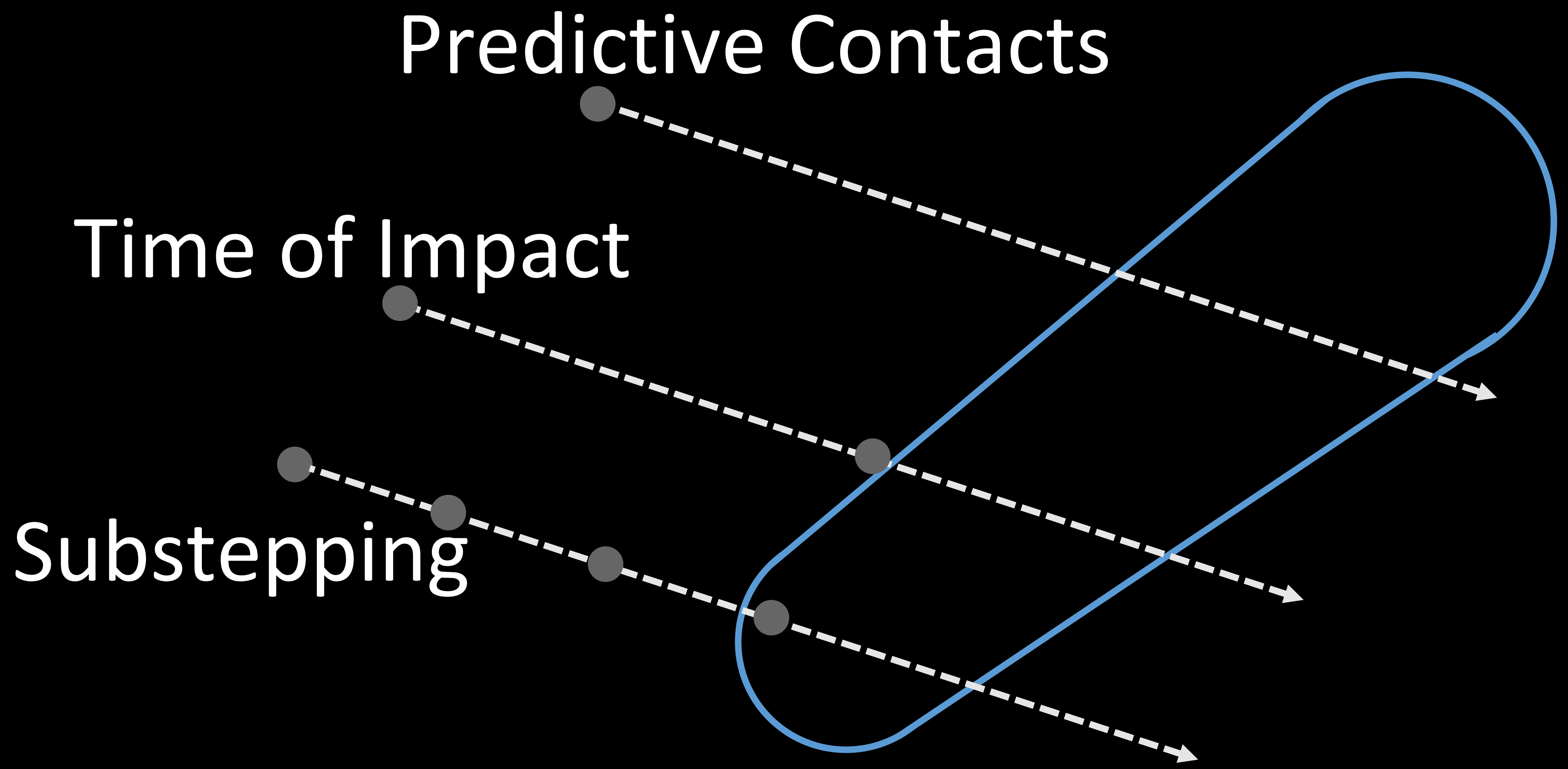
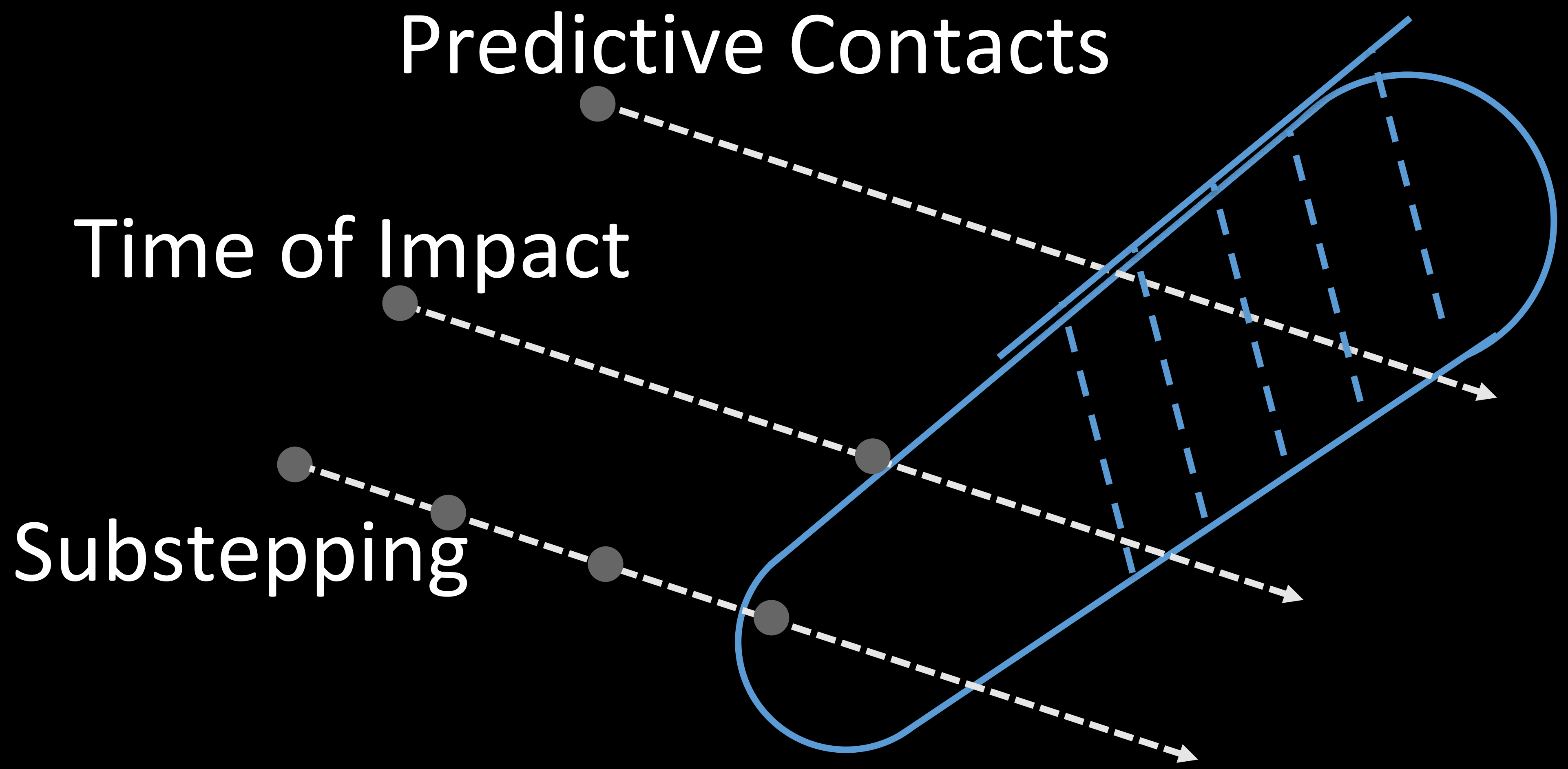


Continuous Collision



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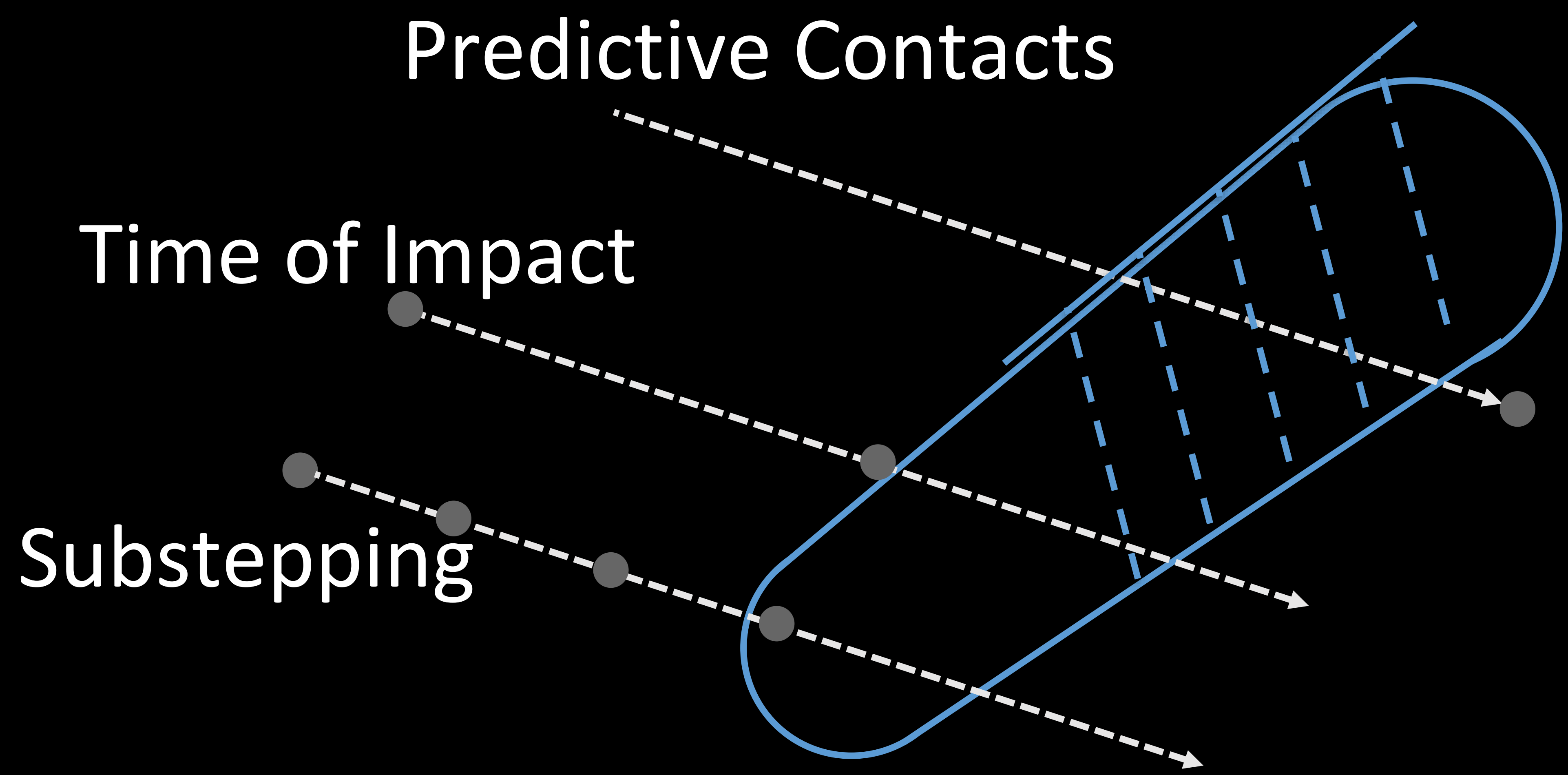

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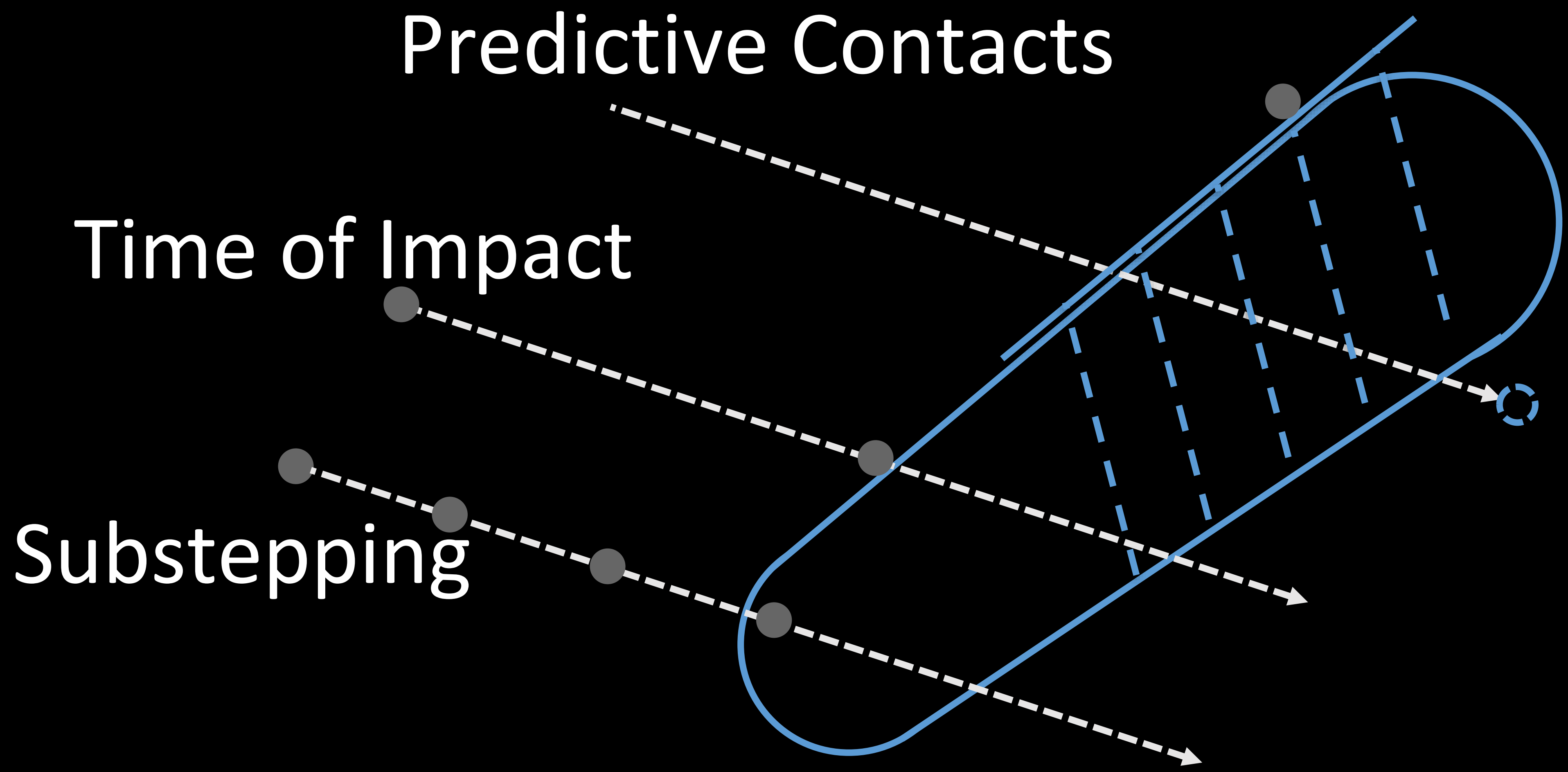

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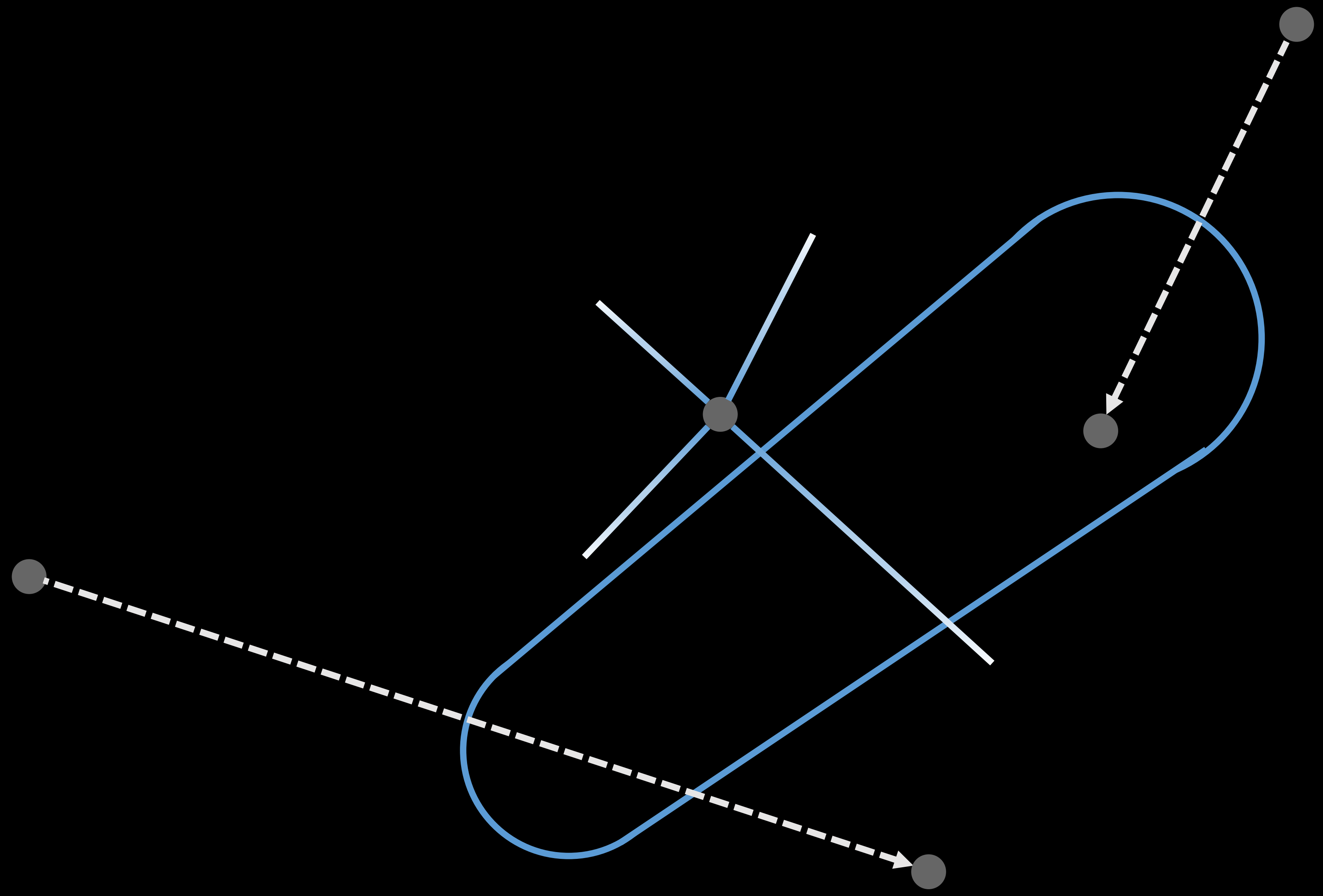

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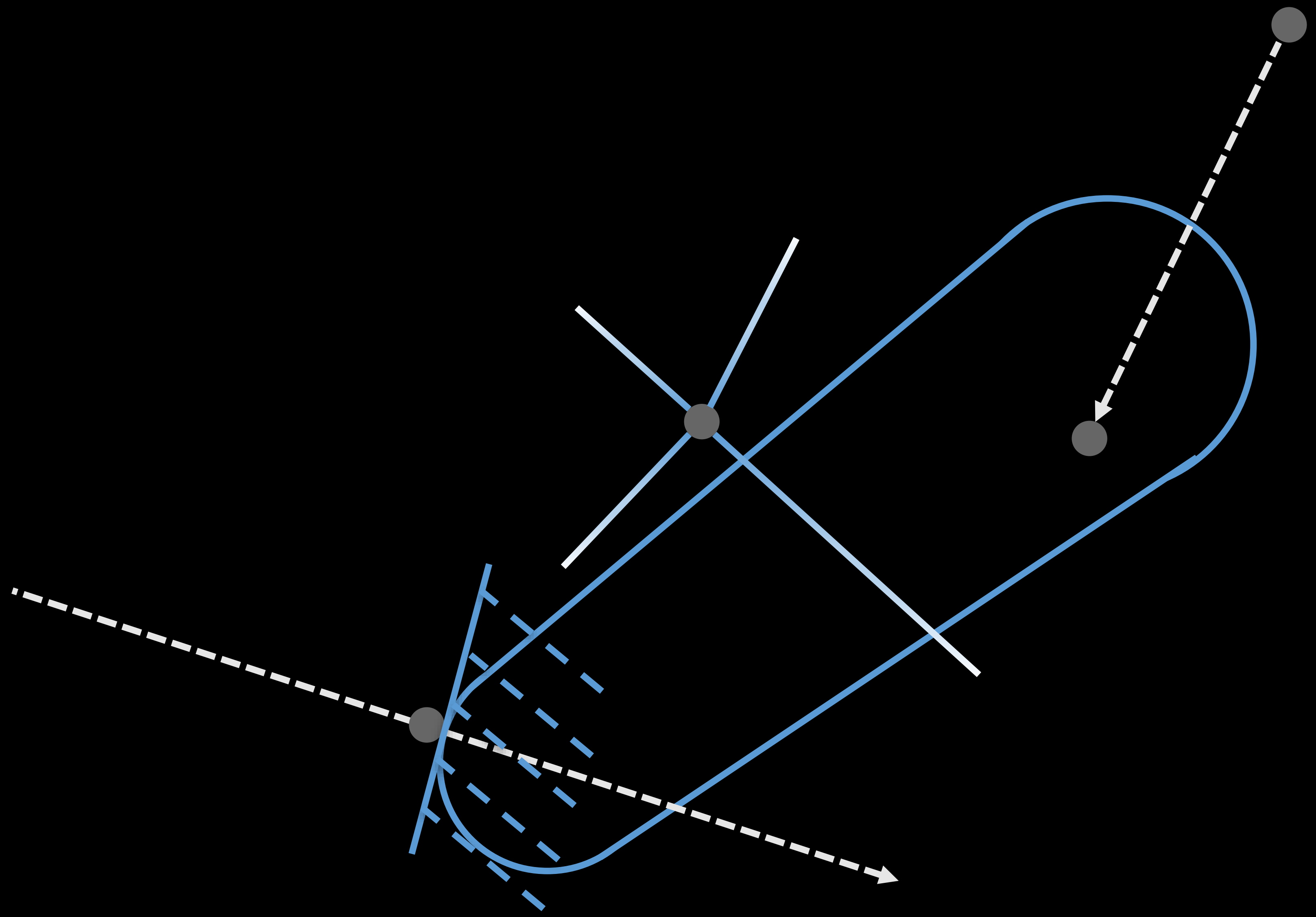

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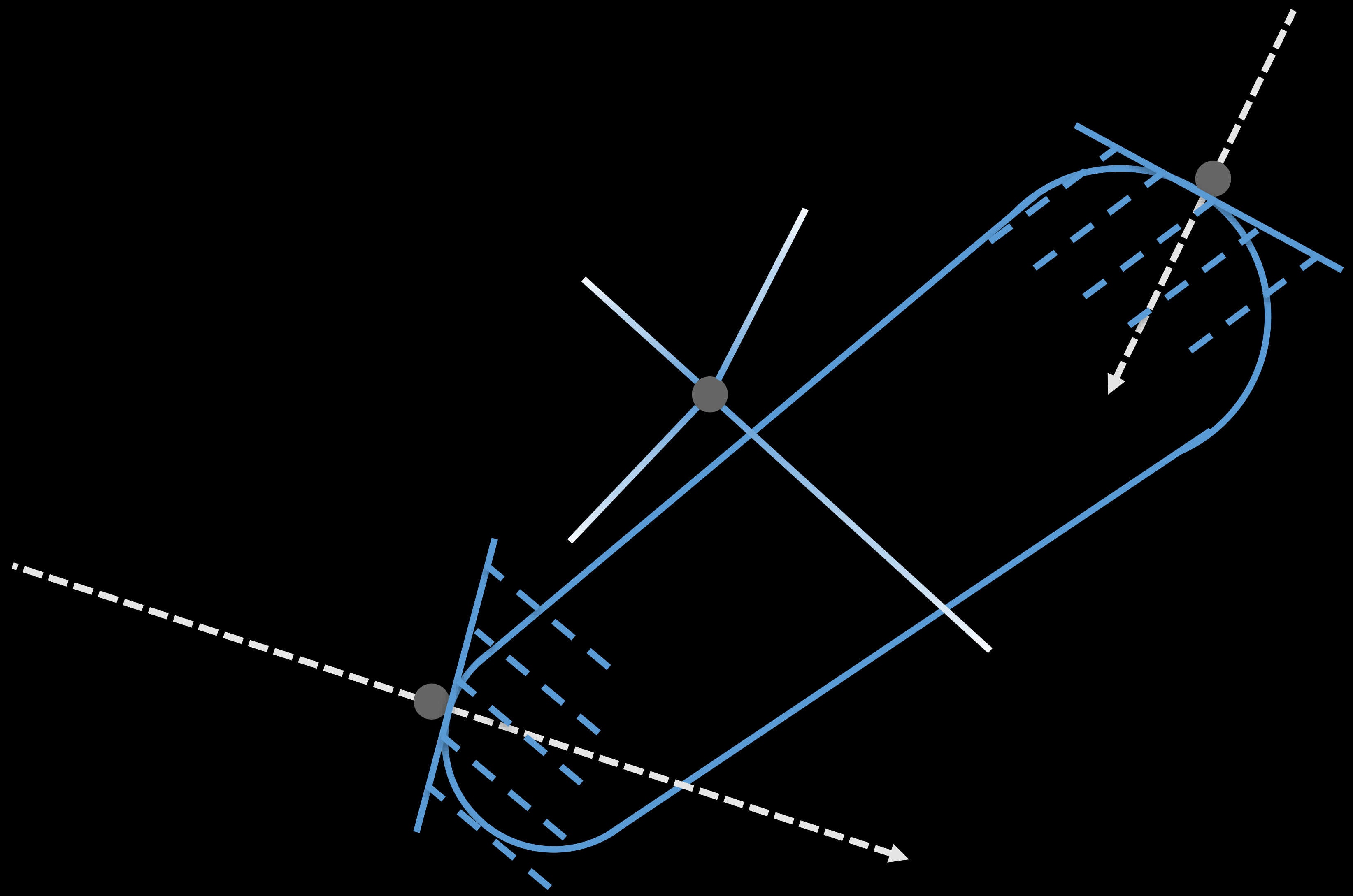
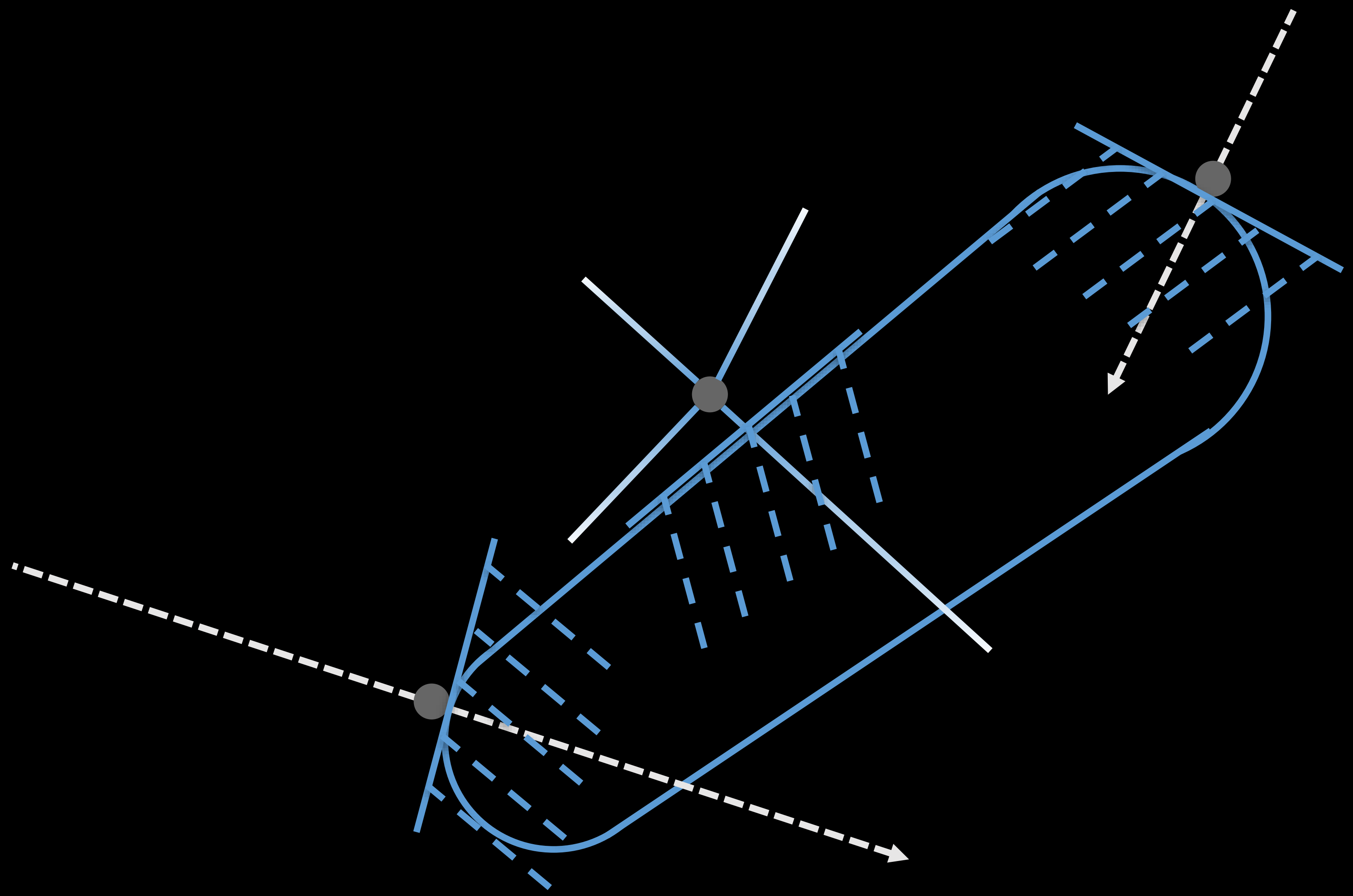


Continuous Collision



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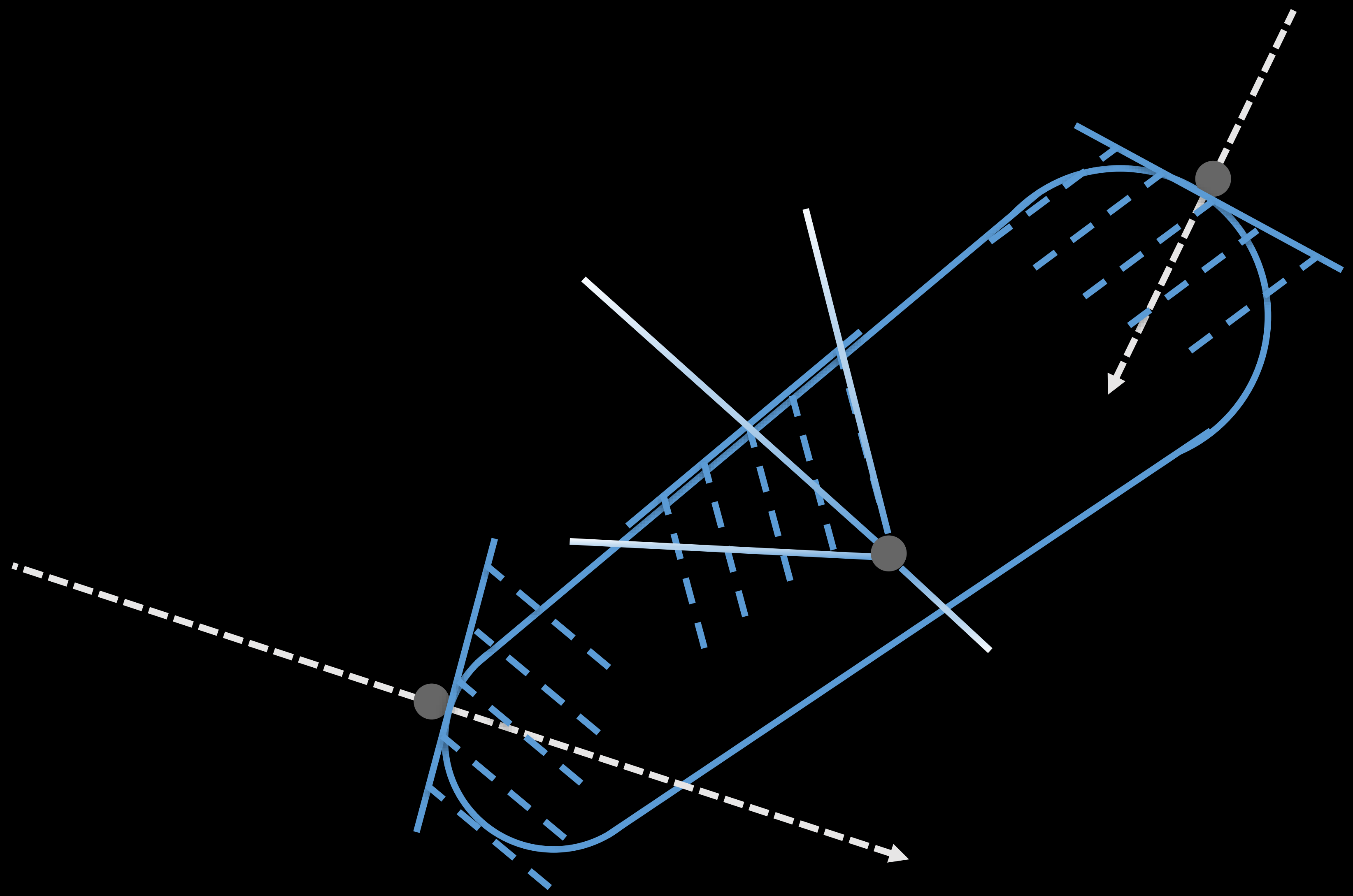

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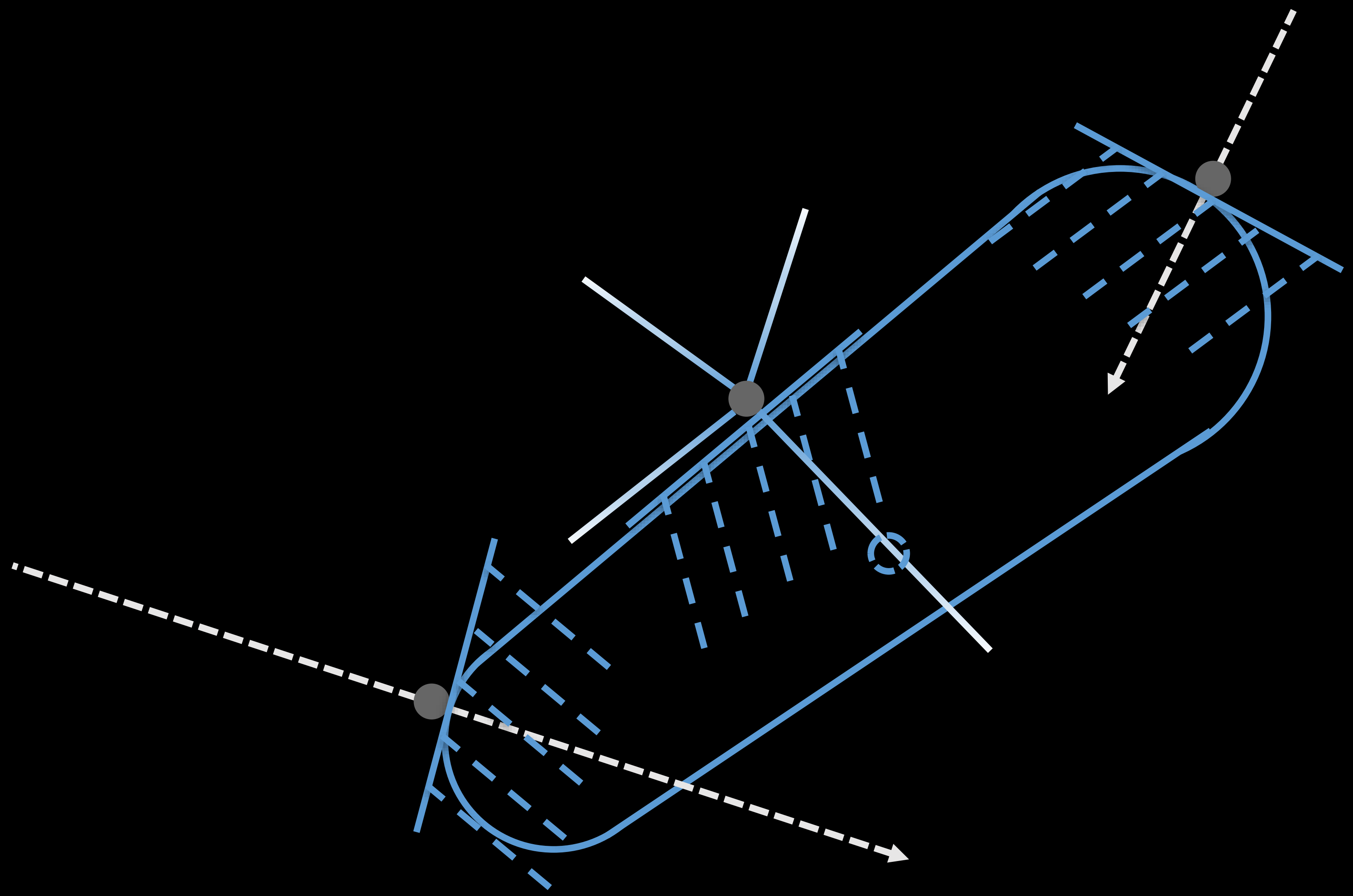

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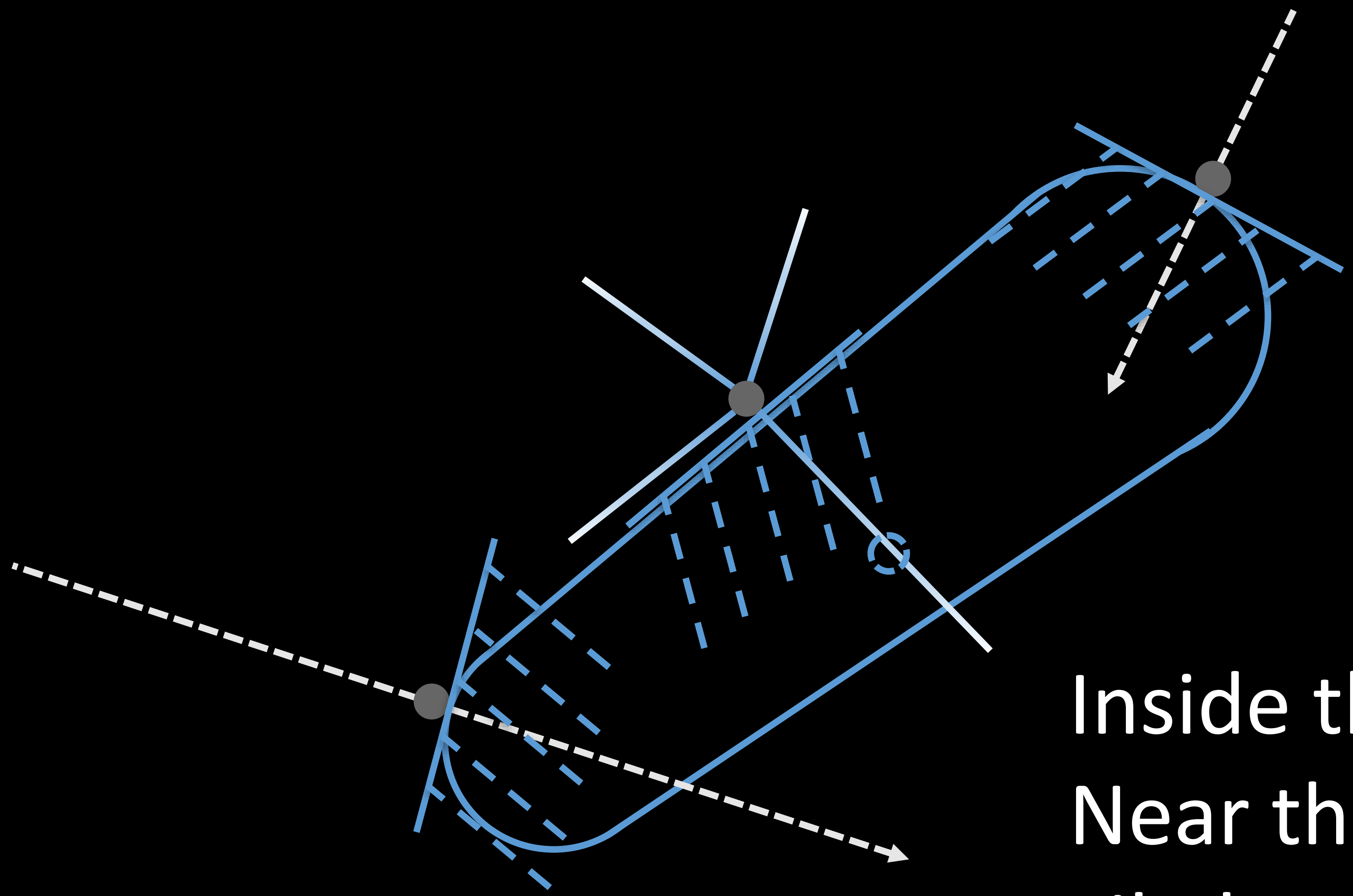
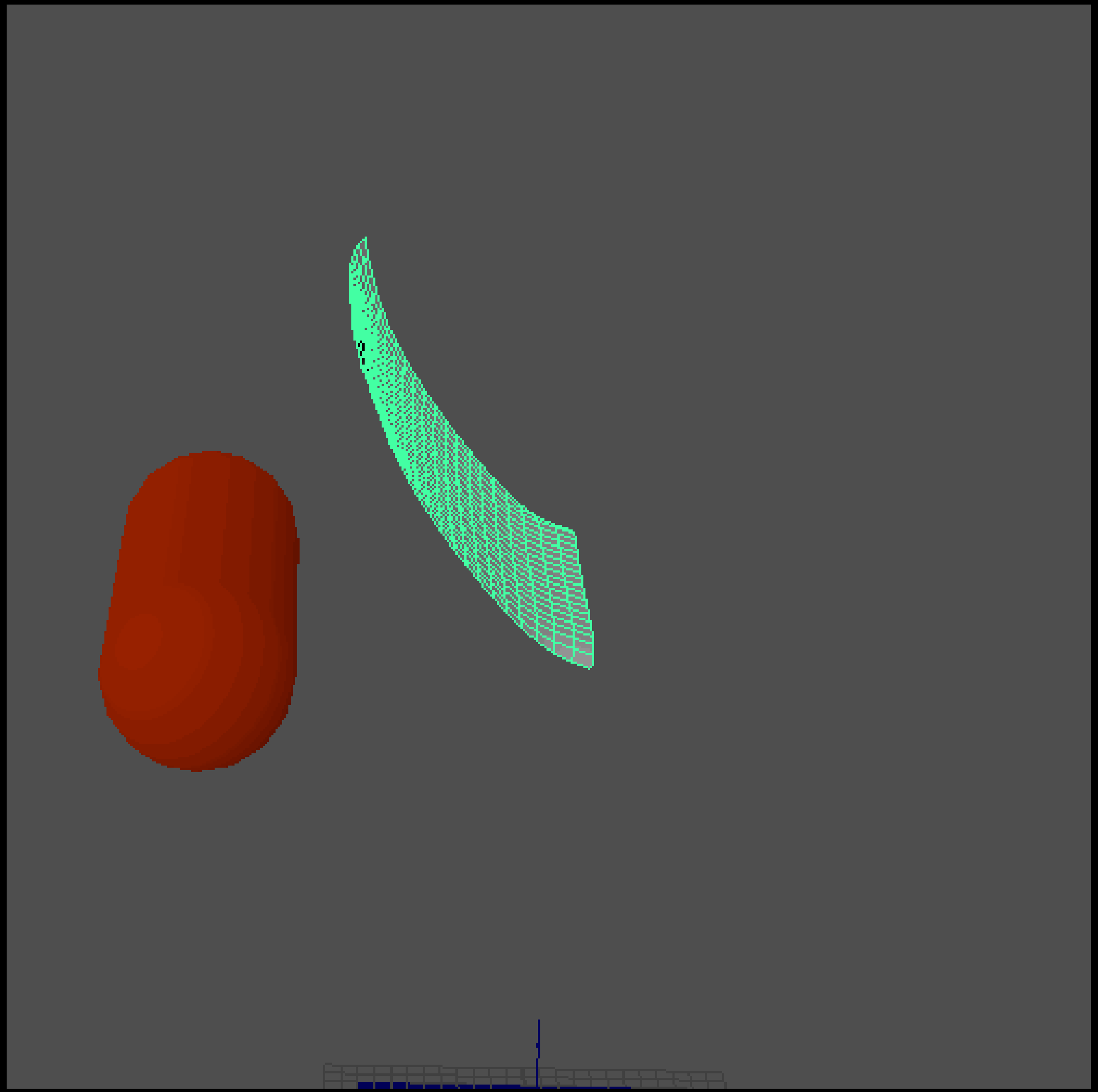


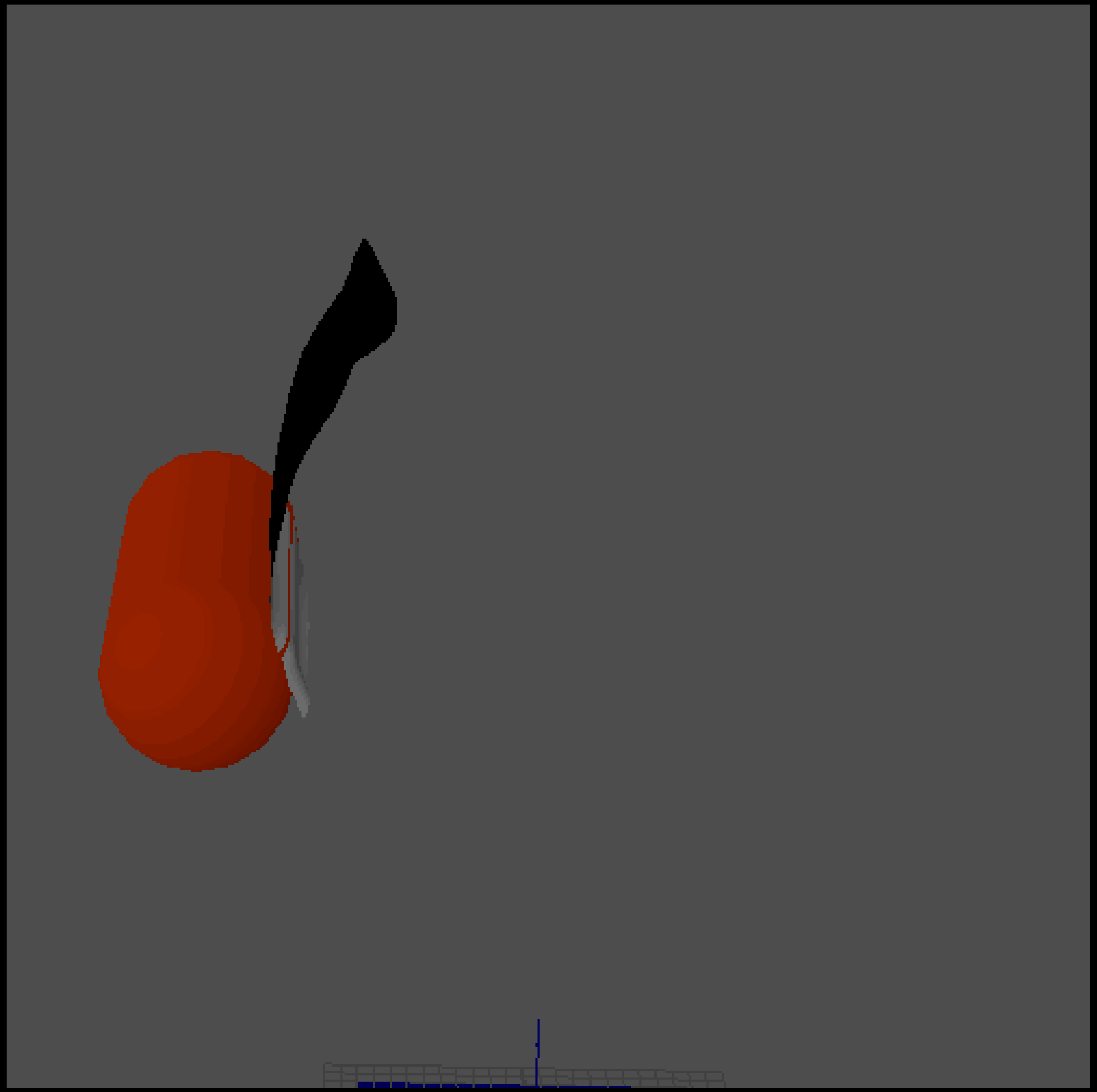
Inside the collider

Near the collider

Likely to be near the collider

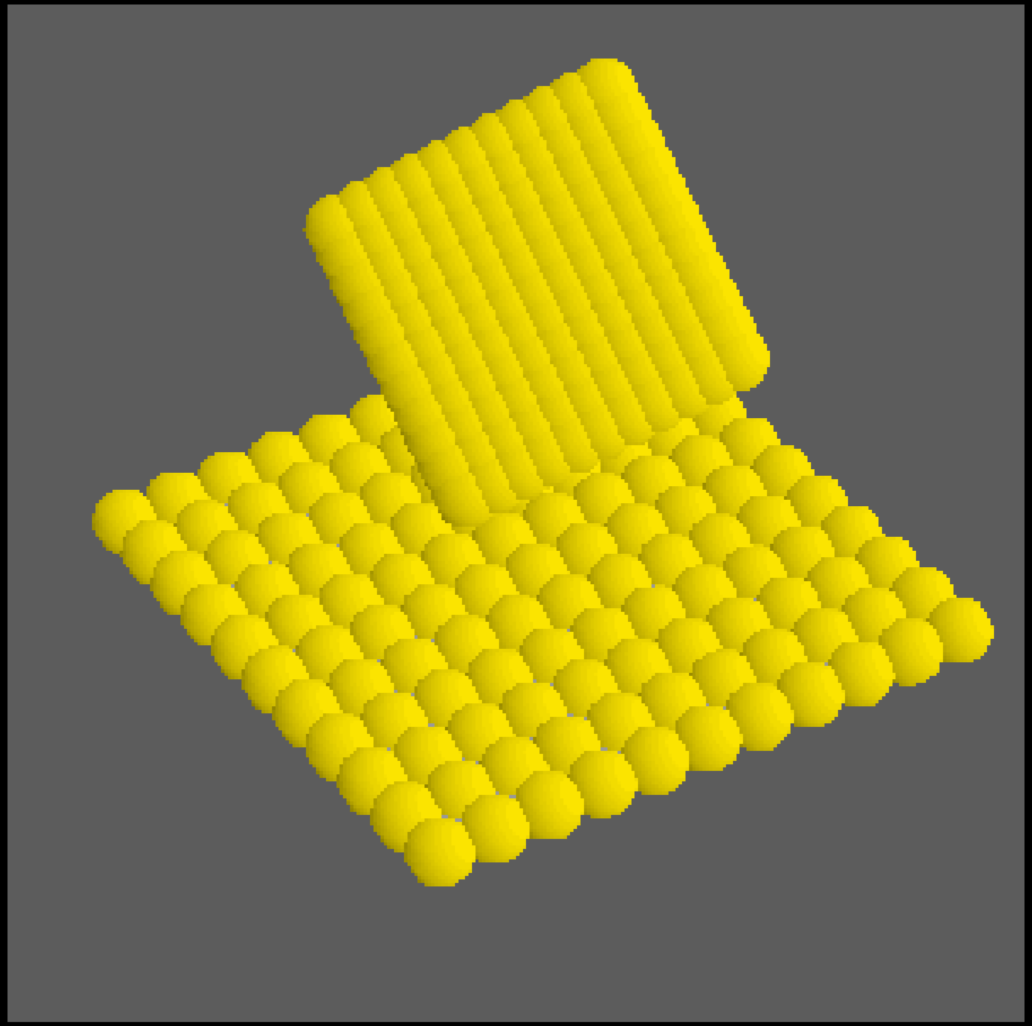


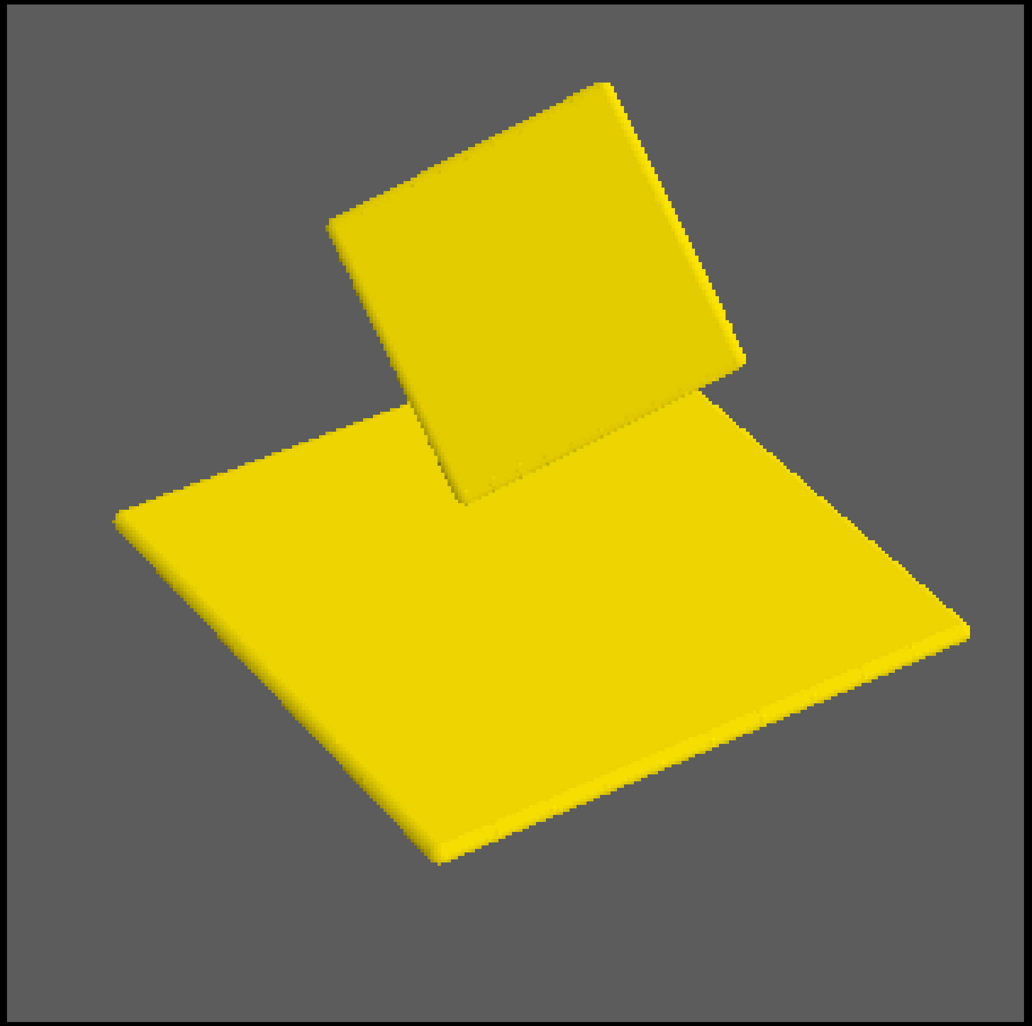
Discrete Collision



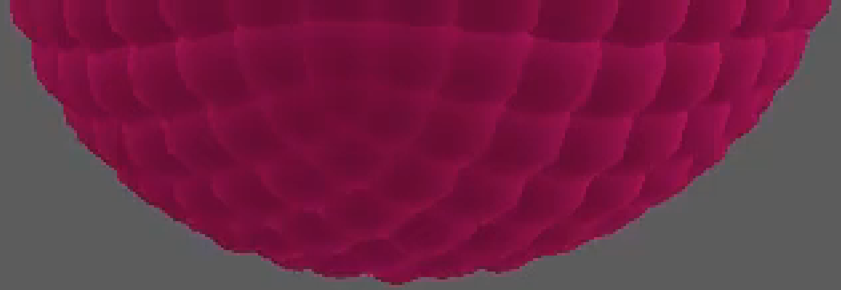
Predictive Contacts

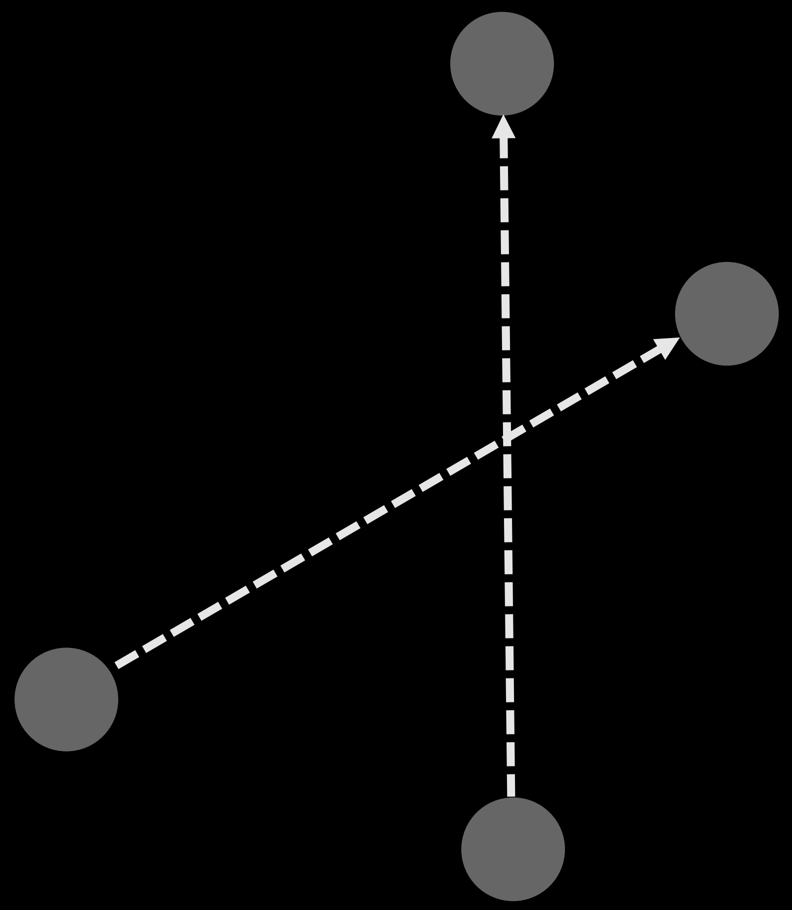
Self Collision



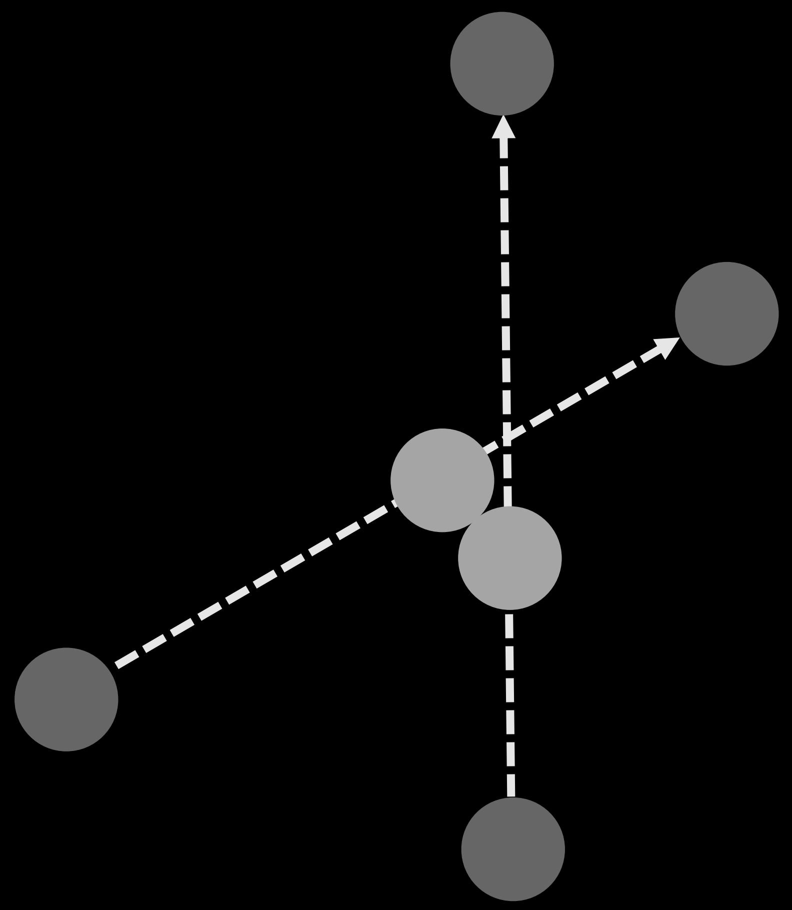


Point-Point Collision

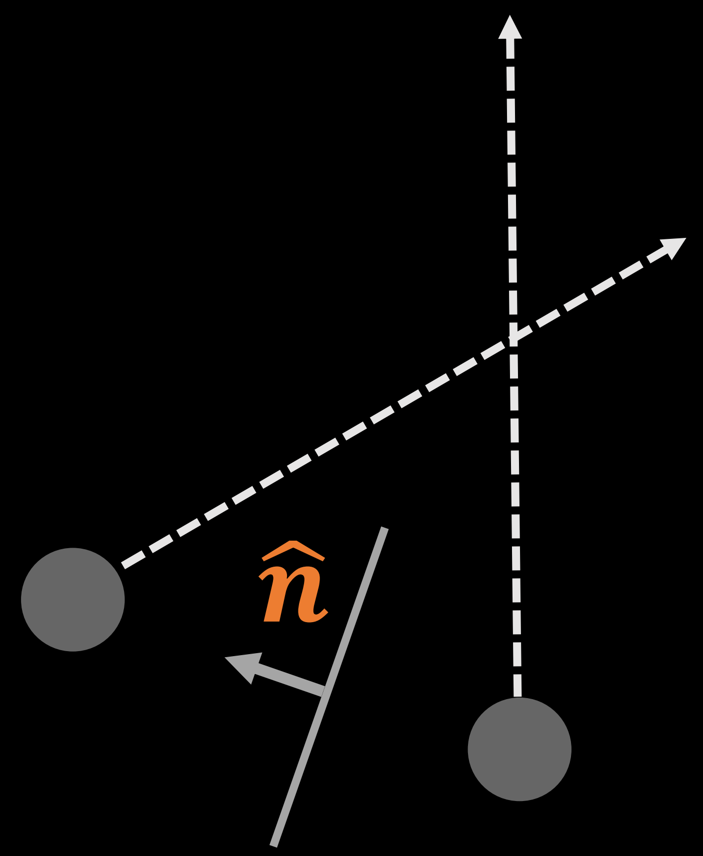




Contact Generation



Contact Generation

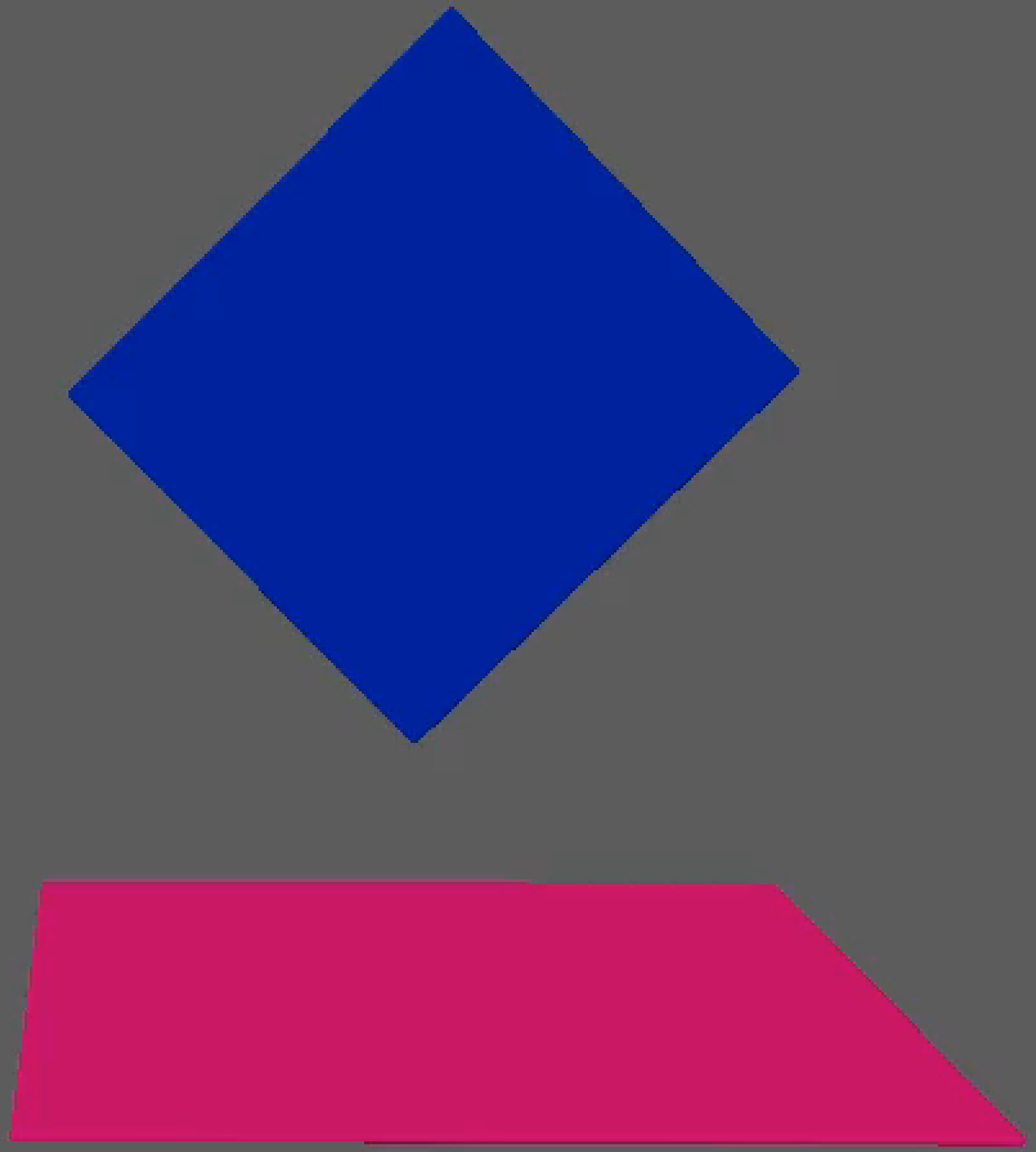



```
// Describes a contact between two points of the
// cloth mesh, for self collision.
struct PointPointContactDesc
{
   // Offset normal between the points at the
   // beginning of the frame.
   Vector3 normal;
   // Index of the 0th point
   uint16_t p0;
   // Index of the 1st point
   uint16_t p1;
};
```

Contact Generation

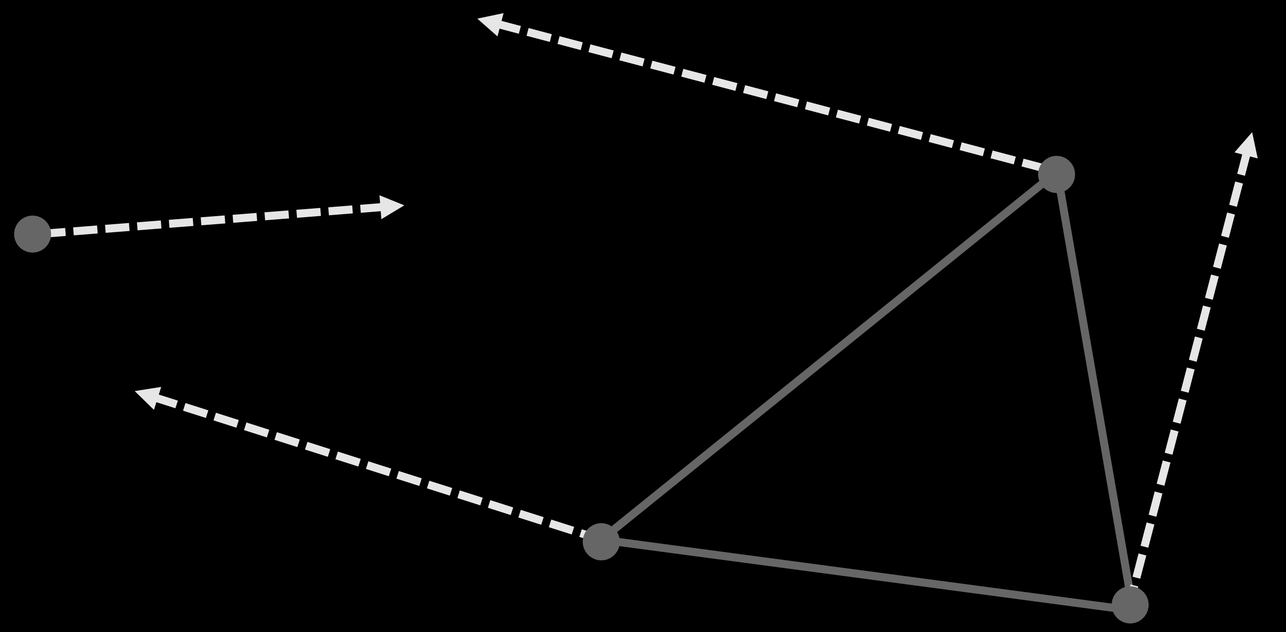
Full Mesh Collision

Edge-Edge

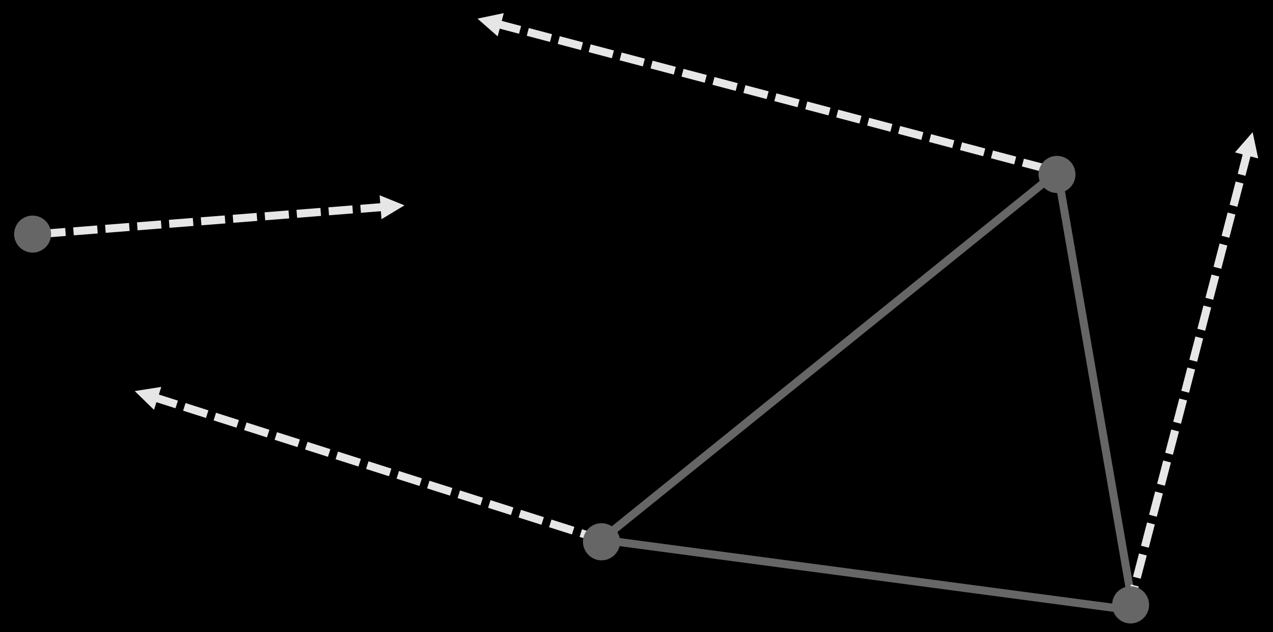


Point-Triangle

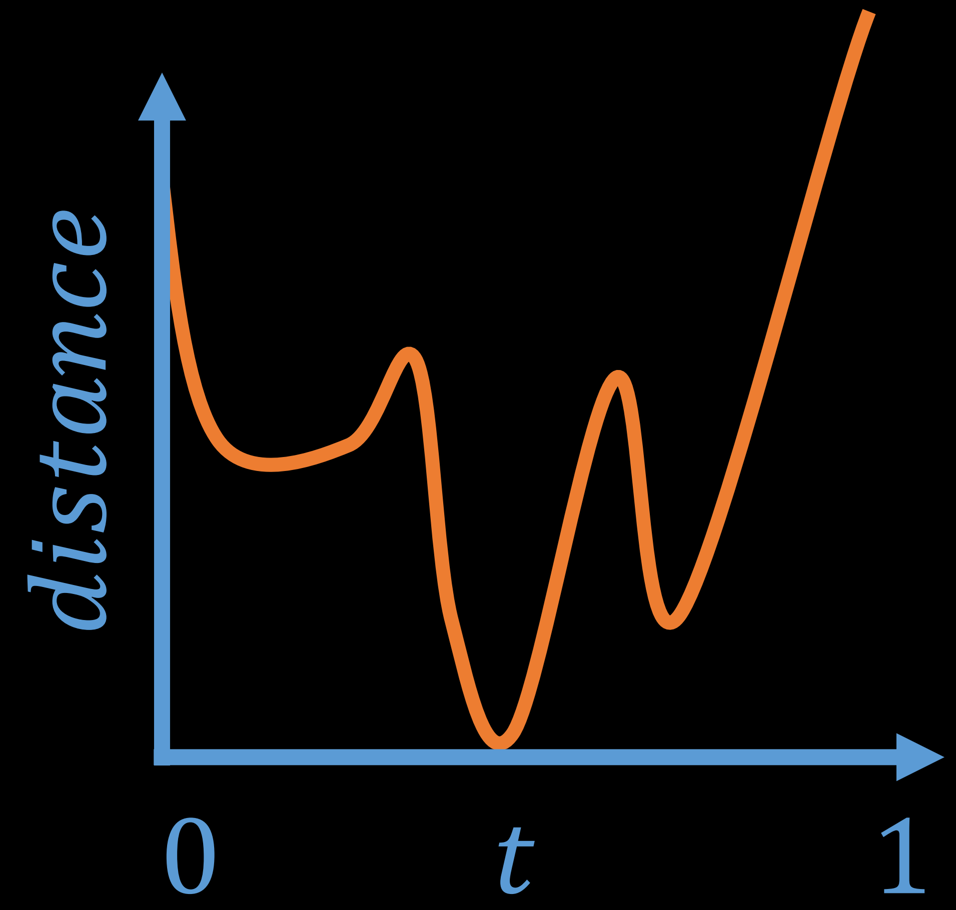
Point-Triangle



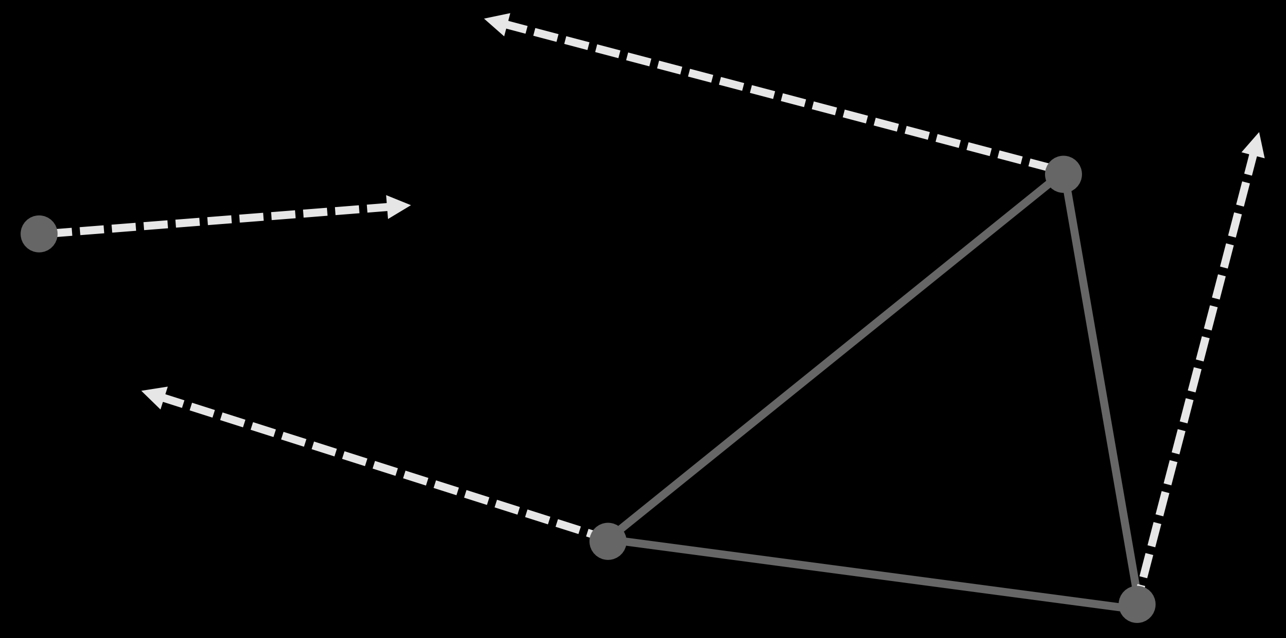
Point-Triangle

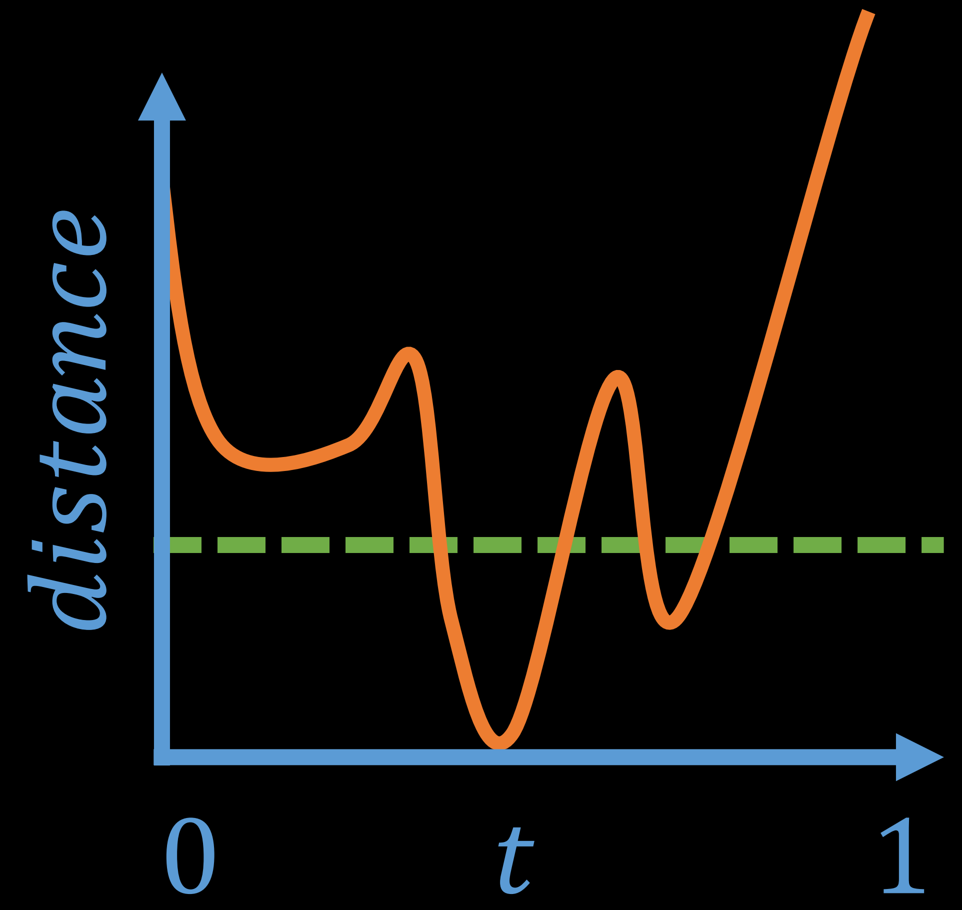


Time of Closest Separation: Sextic
Time of Impact: Cubic



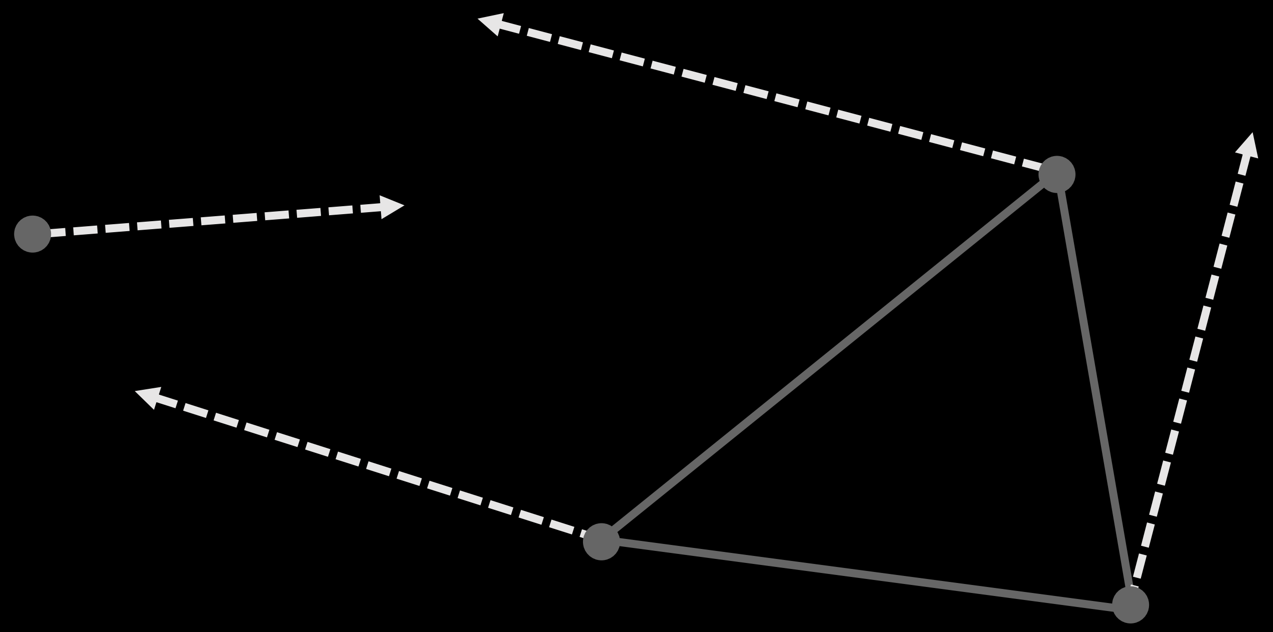
Point-Triangle



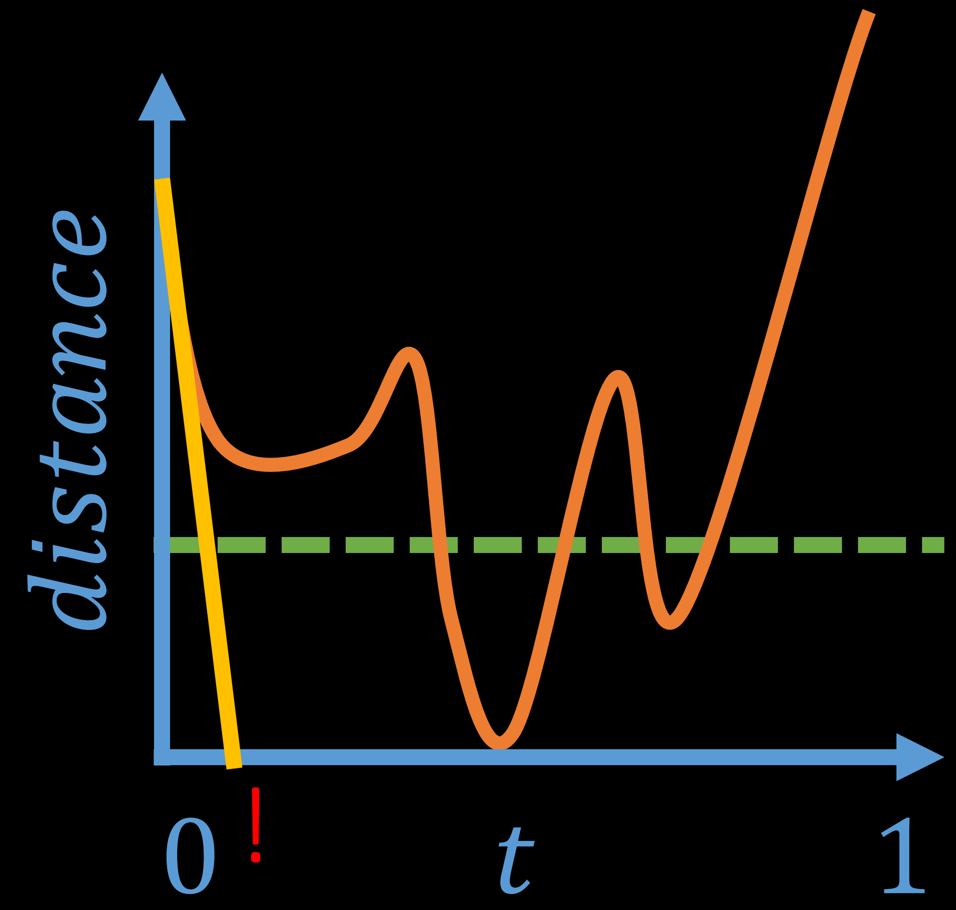


Time of Closest Separation: Sextic
Time of Impact: Cubic

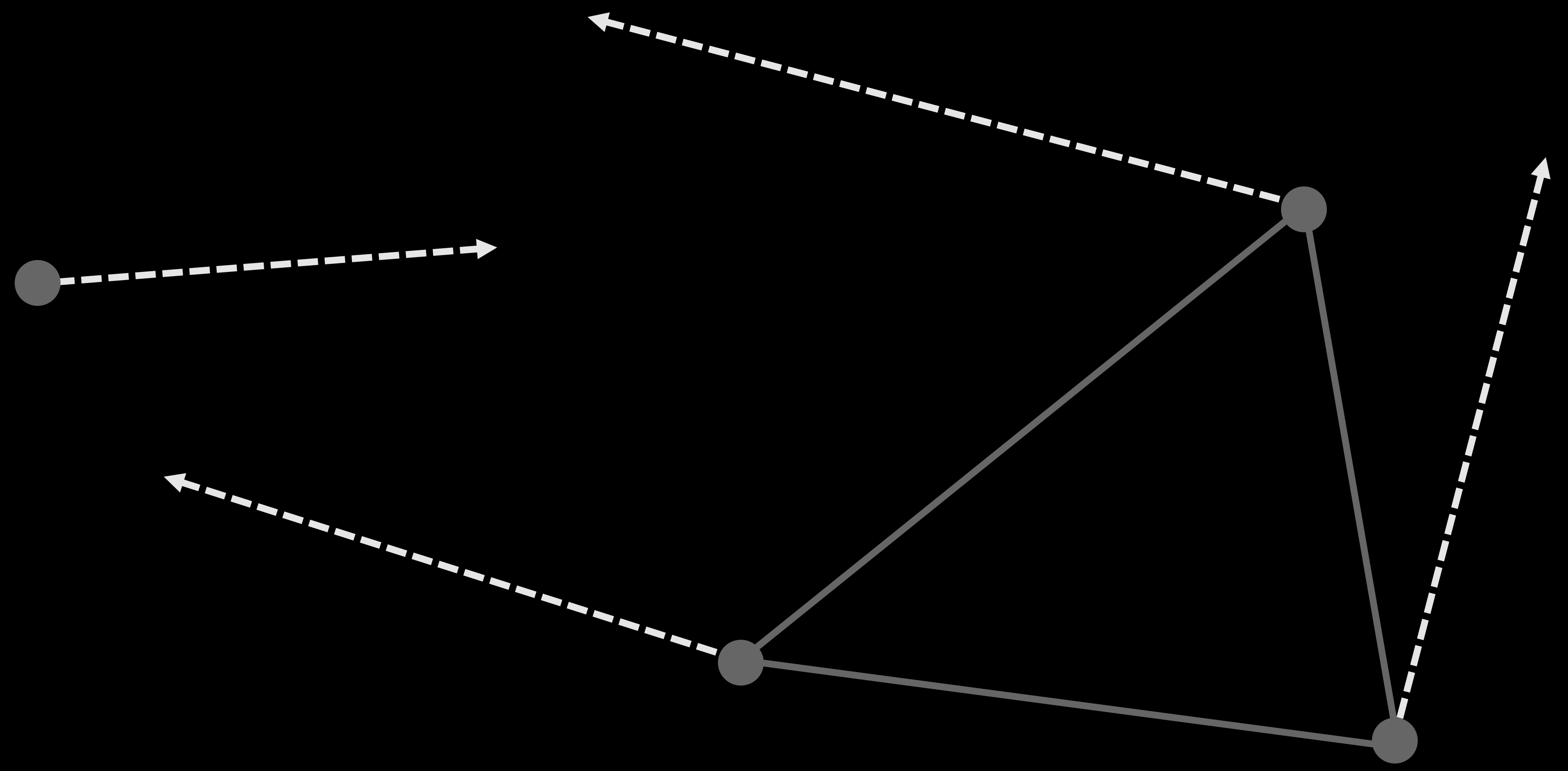
Point-Triangle



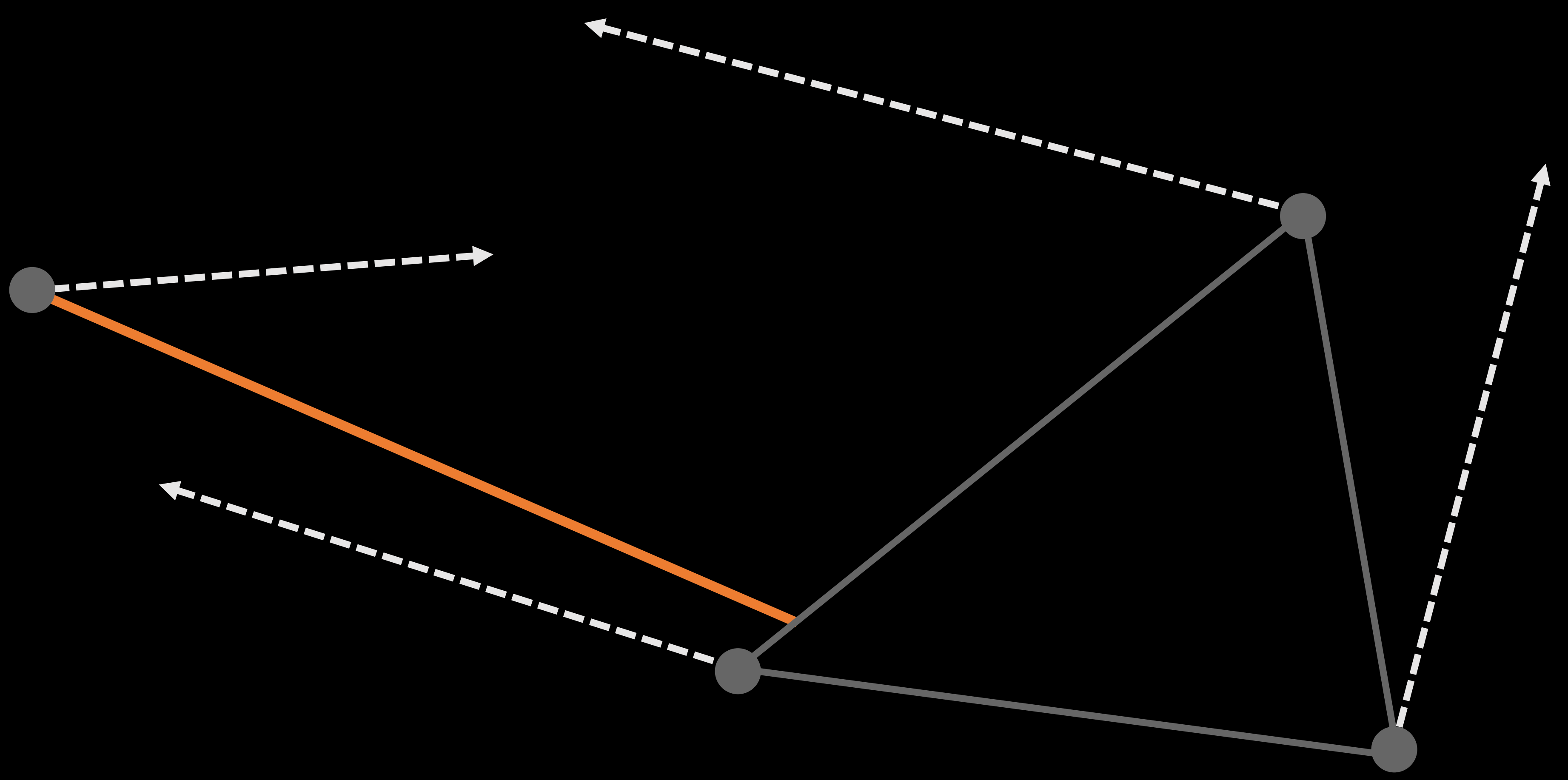
Time of Closest Separation: Sextic
Time of Impact: Cubic
Dynamic Collision Radius: Linear



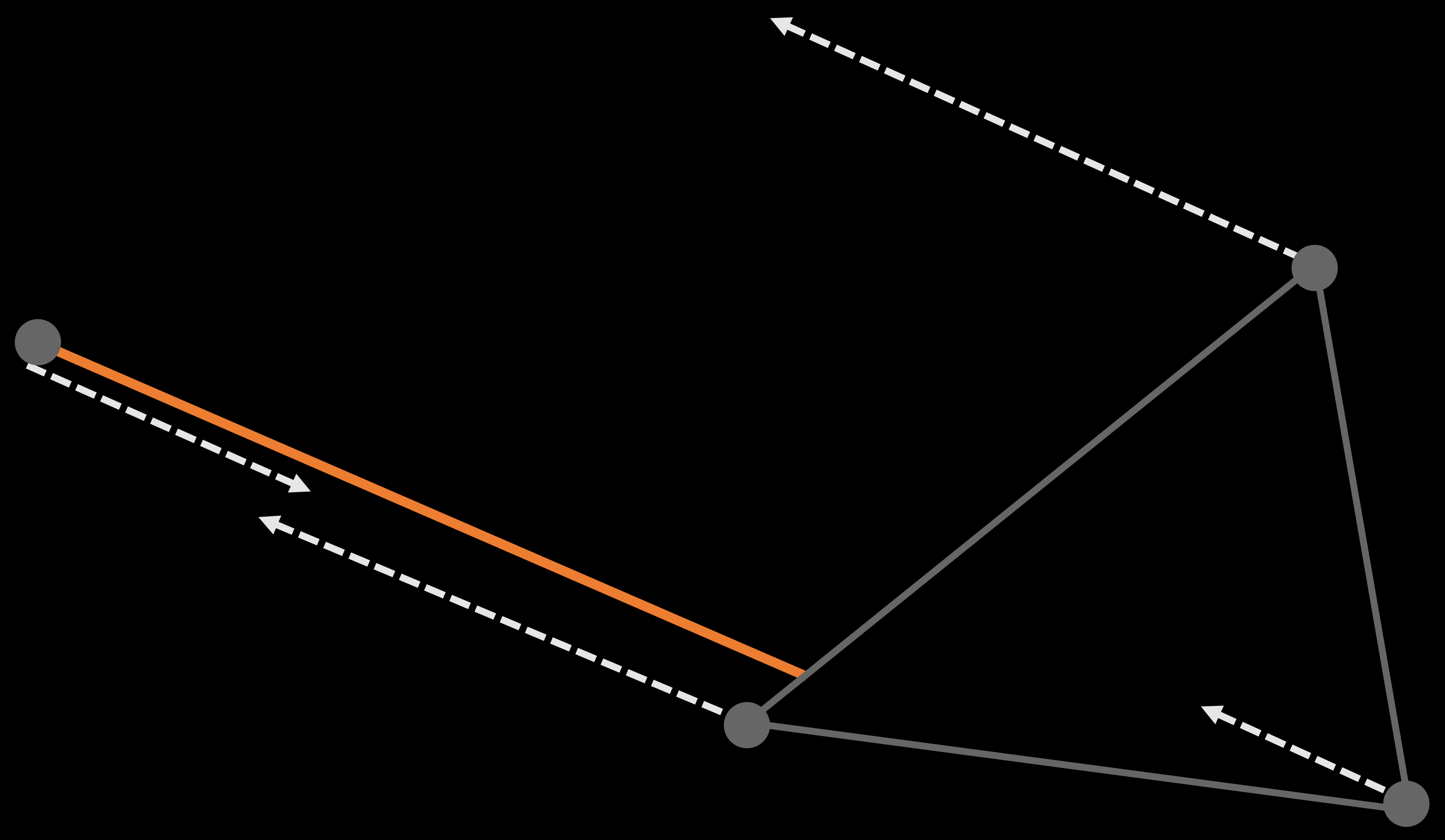
Point-Triangle



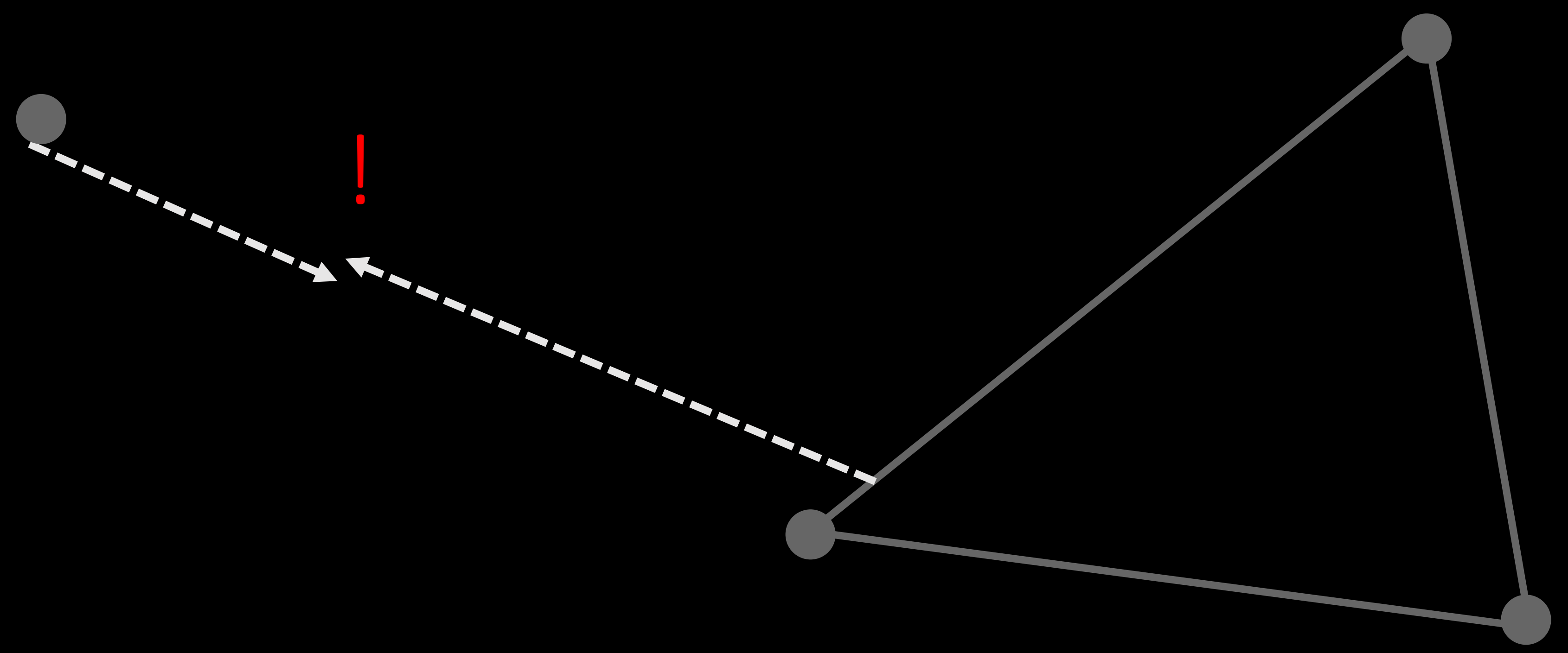
Point-Triangle



Point-Triangle

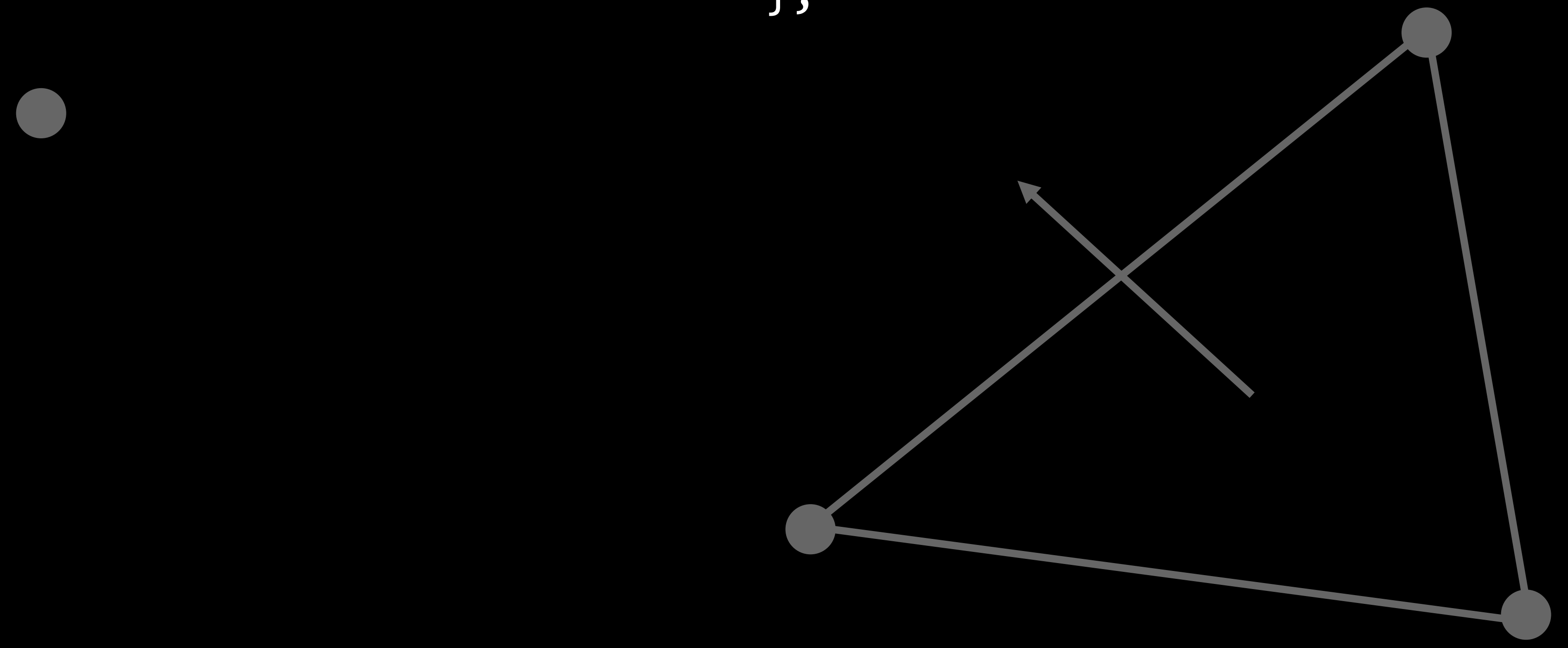


Point-Triangle

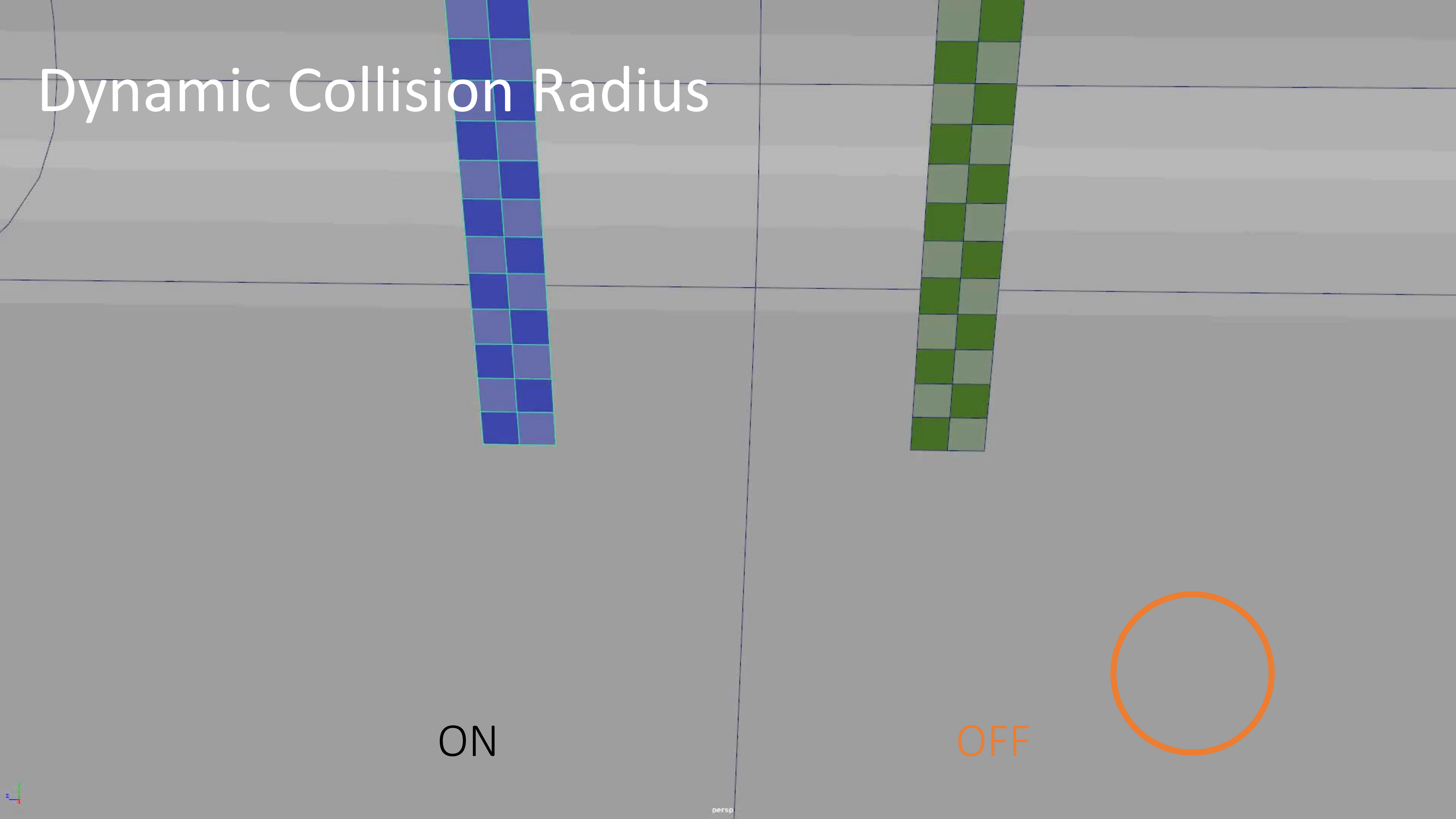


Point-Triangle

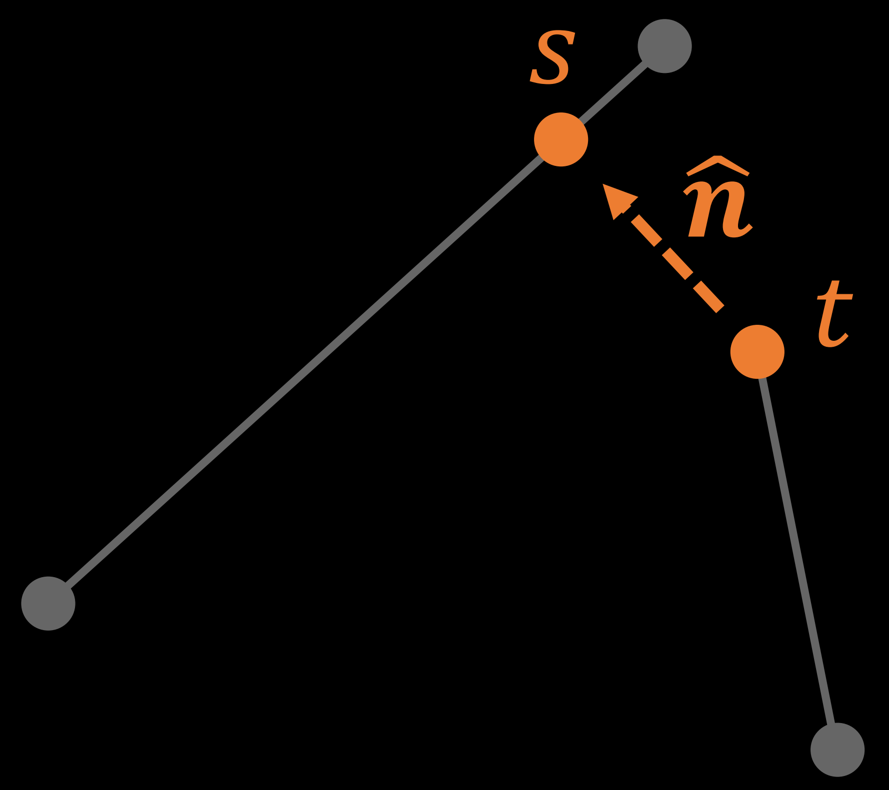
```
// Describes a contact between a cloth point and cloth
// triangle, for self collision.
struct PointTriangleContactDesc
{
    // Index of the point
    uint16_t pointIndex;
    // Index of the triangle.
    uint16_t triangleIndex : 15;
    // Which side of the
    // triangle the point started the frame on.
    uint16_t isBackFace : 1;
};
```



Dynamic Collision Radius

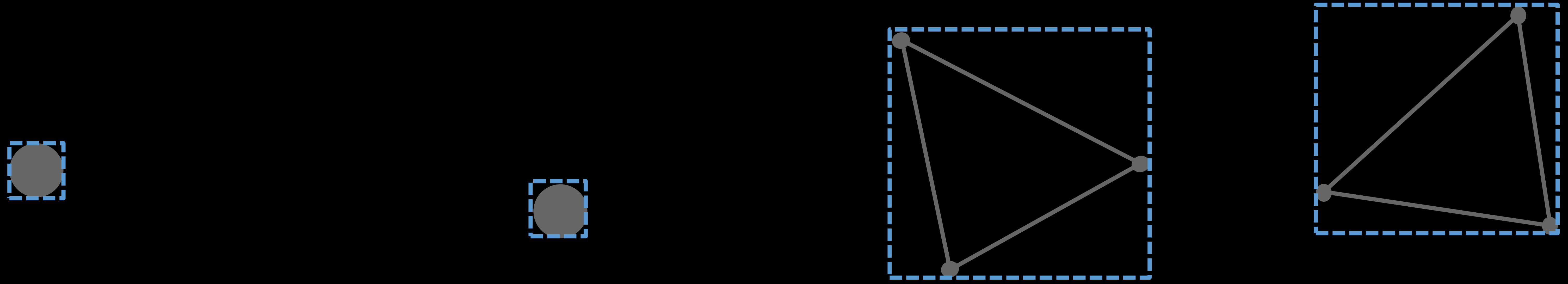


Edge-Edge

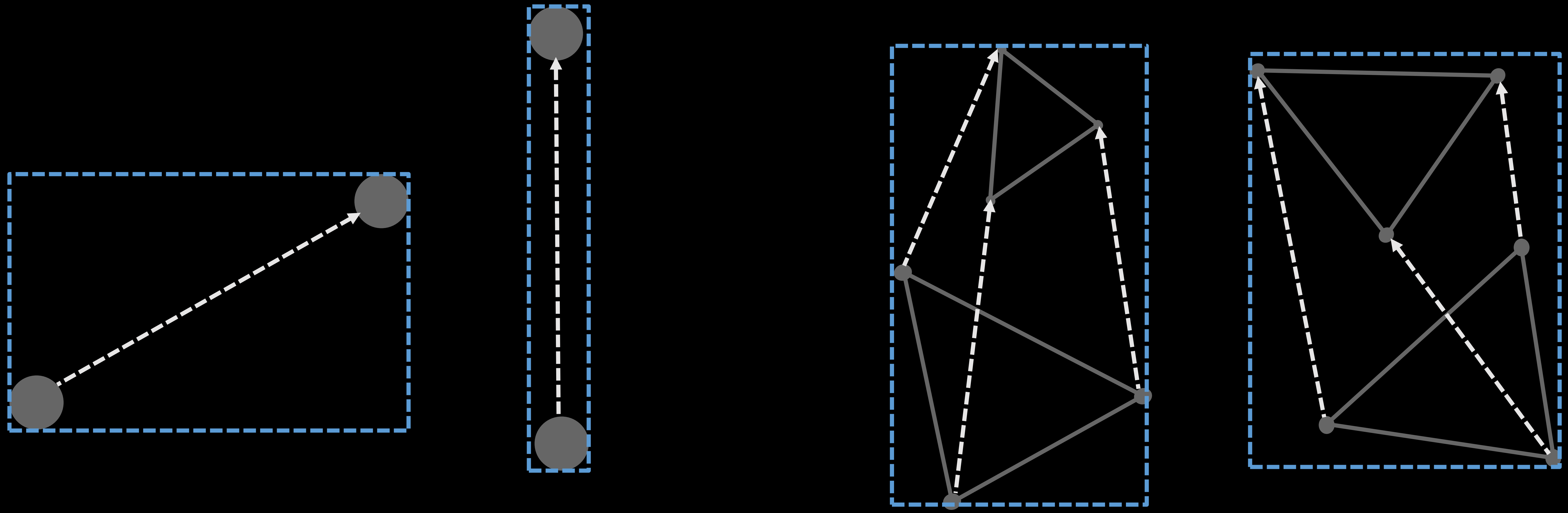



```
// Describes a contact between two edges of the cloth mesh,
// for self collision.
struct EdgeEdgeContactDesc
{
    // Offset normal between closest points on the two lines
    Vector3 normal;
    // parameter of closest point on line 0
    float s;
    // parameter of closest point on line 1
    float t;
    // Index of the 0th point on line 0
    uint16_t e0p0;
    // Index of the 1st point on line 0
    uint16_t e0p1;
    // Index of the 0th point on line 1
    uint16_t e1p0;
    // Index of the 1st point on line 1
    uint16_t e1p1;
};
```

Broadphase

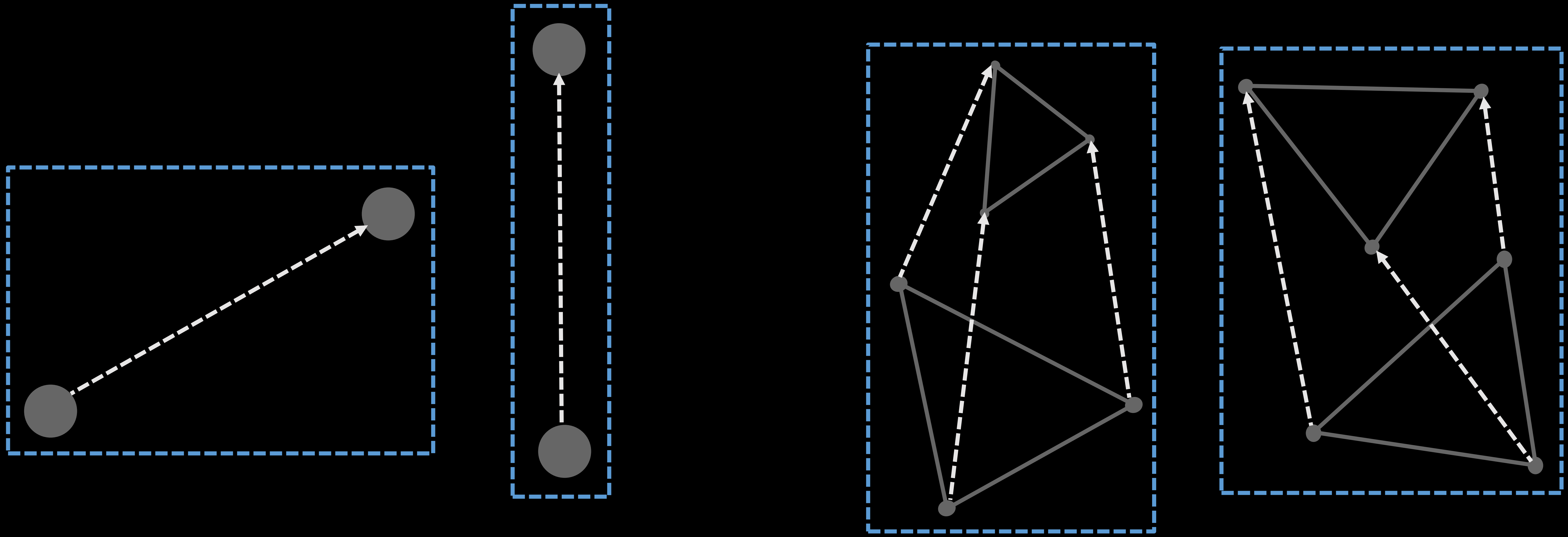


Broadphase

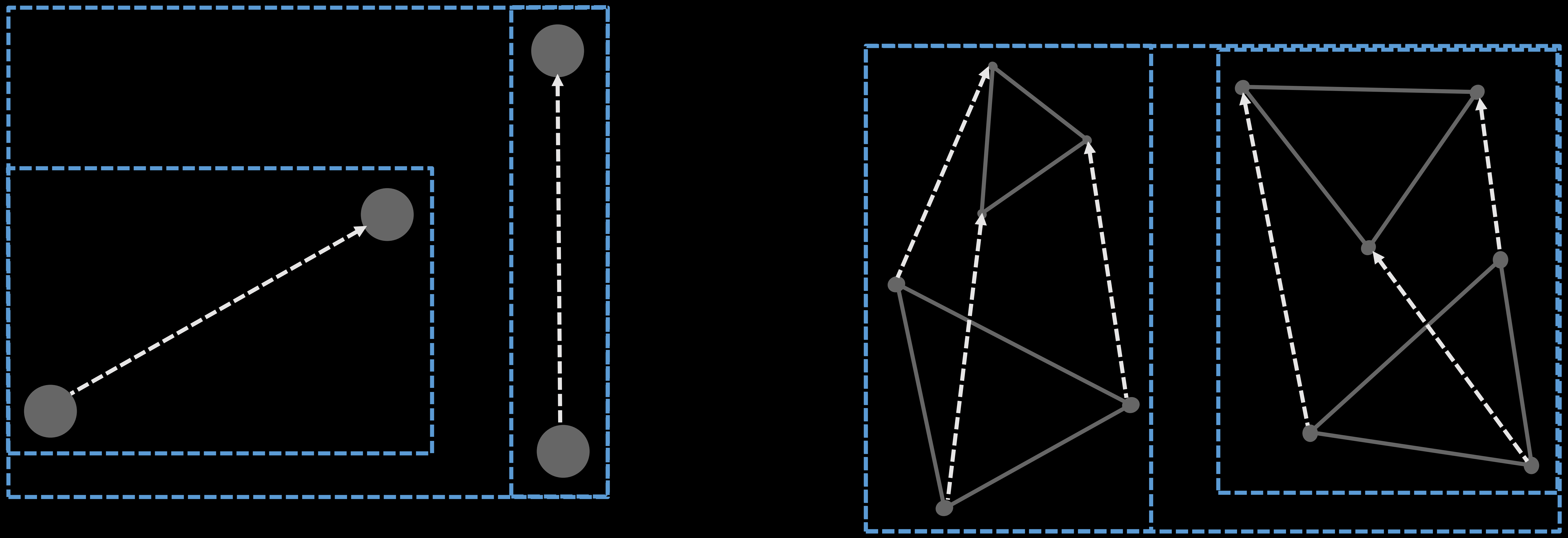


[Ericson, Real Time Collision Detection]

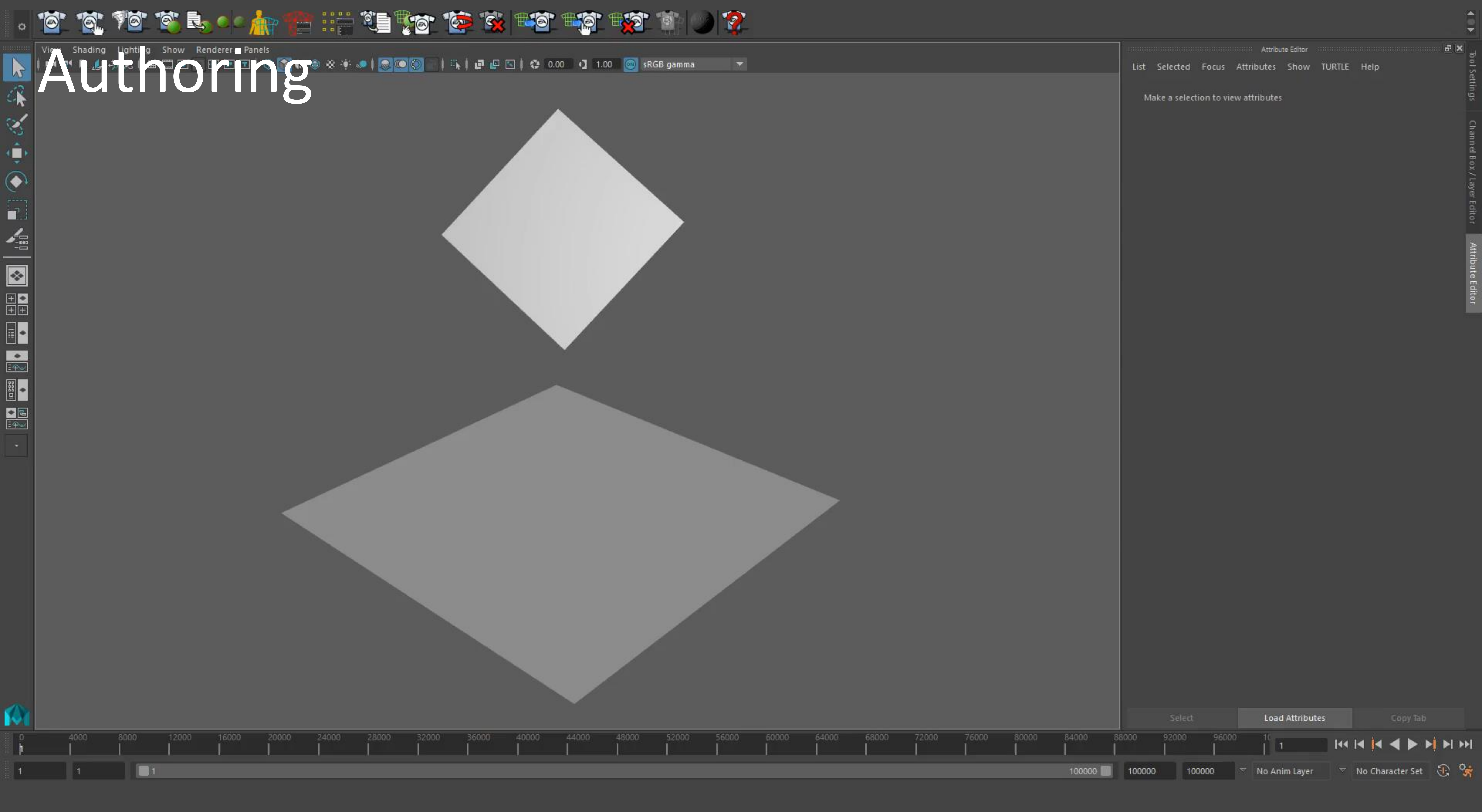
Broadphase



Broadphase



Authoring


View Shading Lighting Show Renderer Panels
0.00
1.00
sRGB gamma
Attribute Editor
List Selected Focus Attributes Show TURTLE Help
Make a selection to view attributes
Select
Load Attributes
Copy Tab
Tool Settings
Channel Box / Layer Editor
Attribute Editor
No Anim Layer
No Character Set

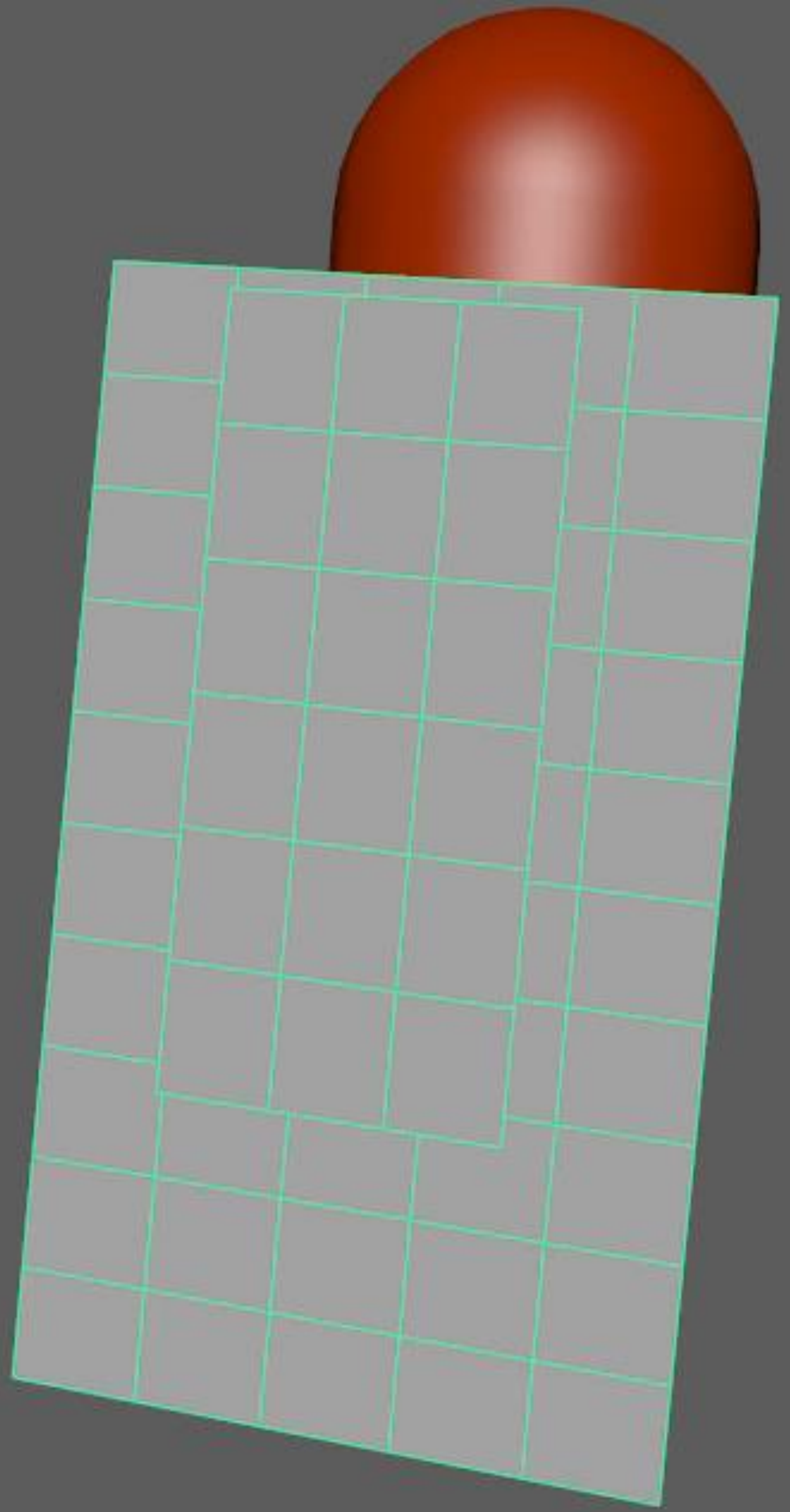
Results

Triangle Collision
1.8k triangles
~20ms / frame
30Hz simulation
1 thread

Results

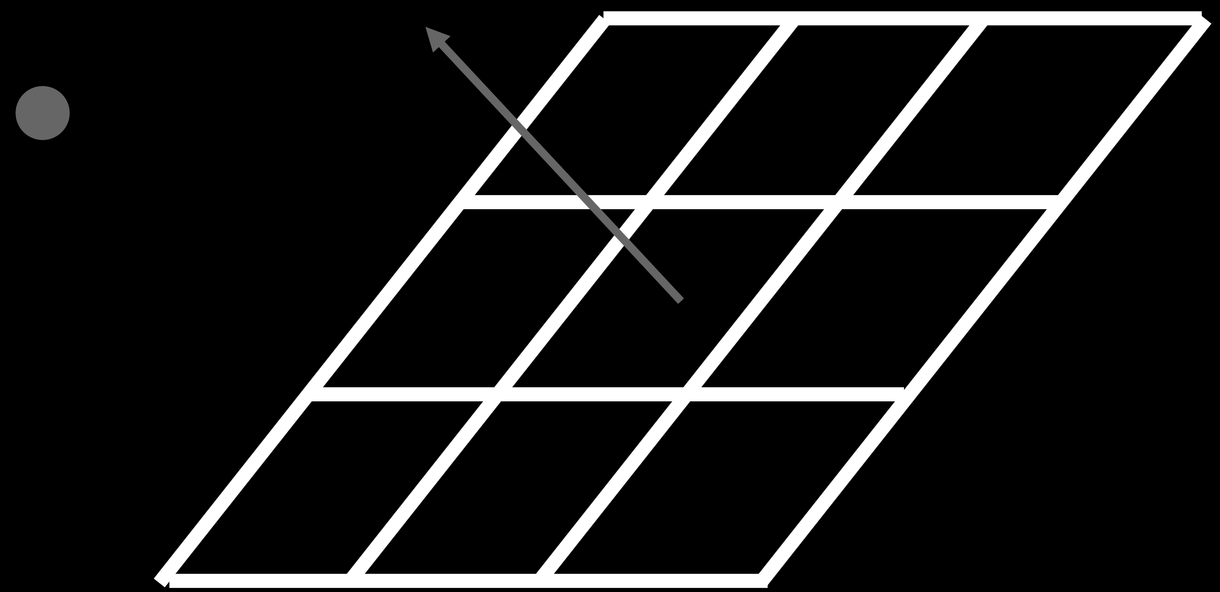
Triangle Collision
28k triangles
~1s / frame
60Hz simulation
1 thread

Results

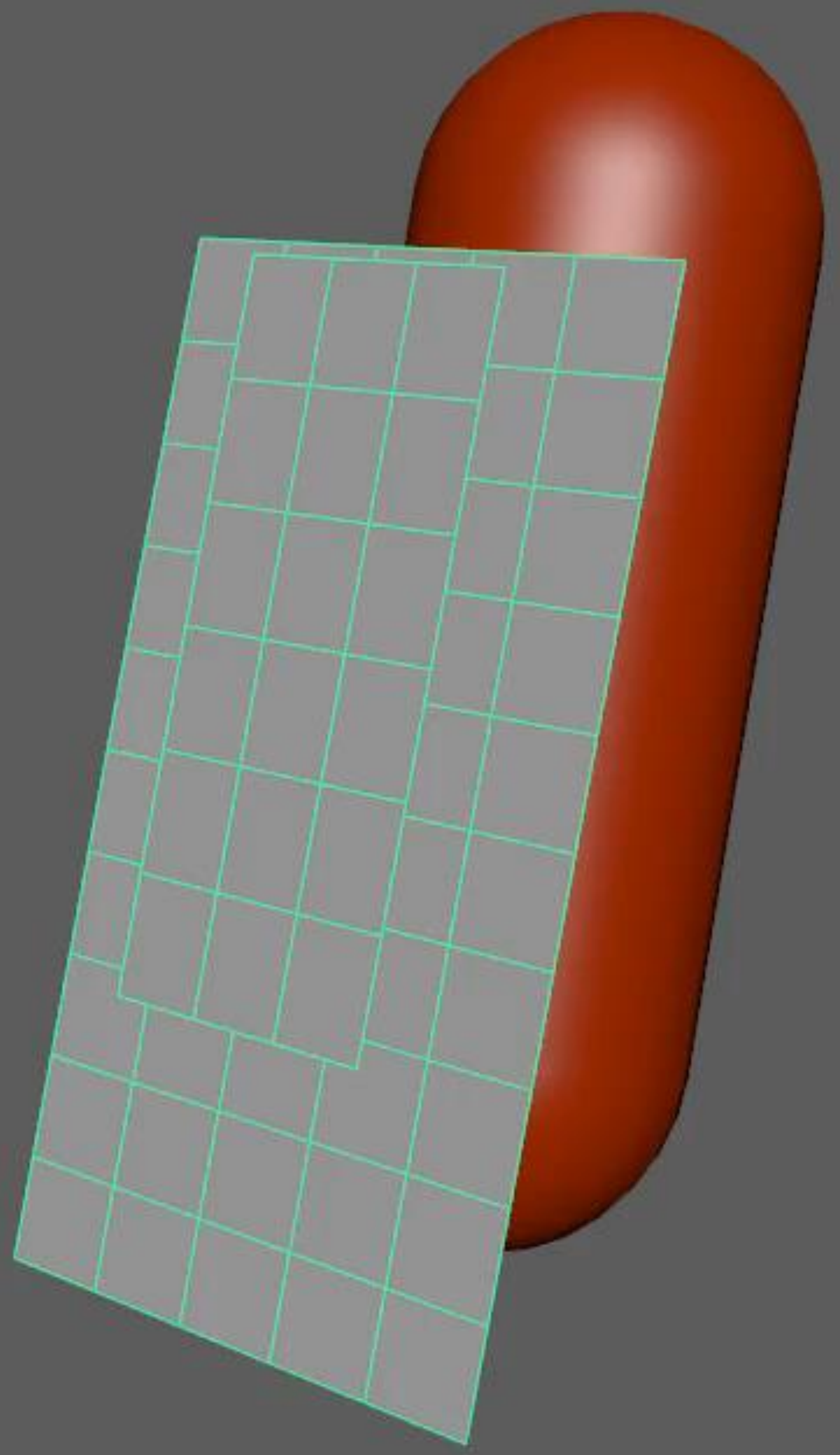


Dirty Tricks - Layered Clothing

Sometimes the surface has a natural orientation.



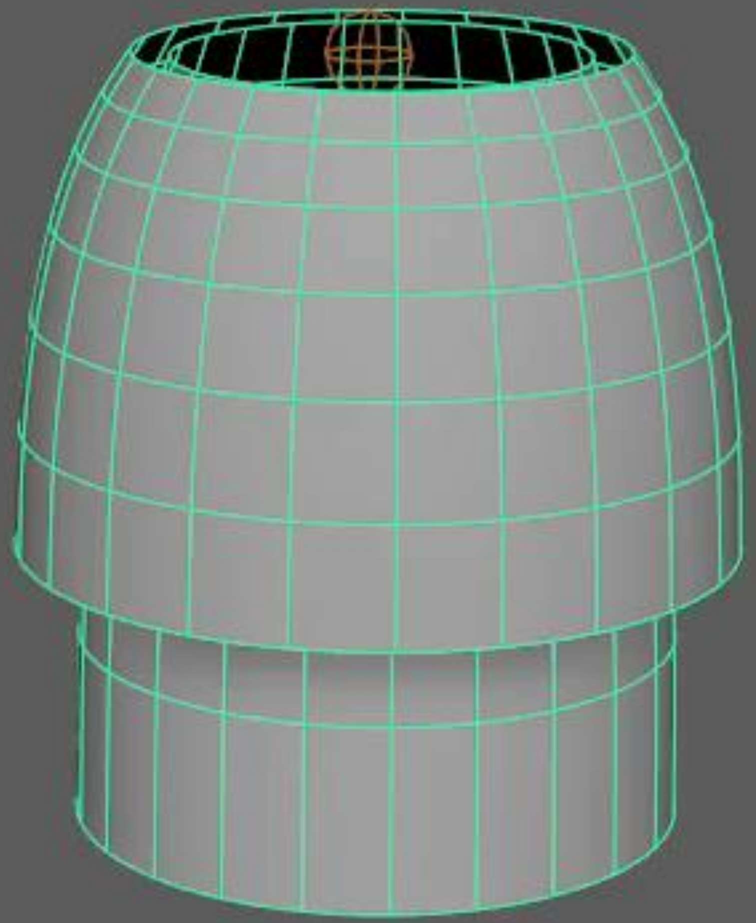
Results



Triangle Collision
0.2ms / frame
60Hz simulation
1 thread

Results

Triangle Collision
3ms / frame
30Hz simulation
1 thread

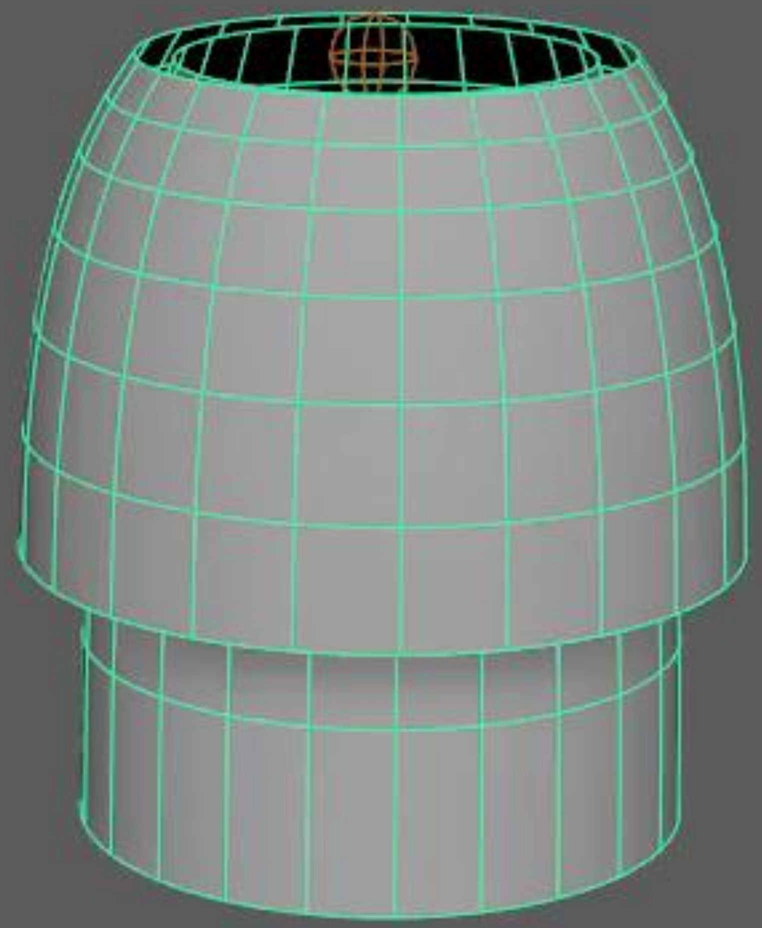


Dirty Tricks – Contact Caching

What if we only do the collision detection once?

Just keep the same contact set every frame

Contact Caching



3ms / frame -> 0.2ms / frame!

Performance in this example

```
380µs Total solve
      60 µs Collision (points vs. colliders)
      150µs Self-collision constraints
      150µs All other constraints
```

Intel desktop CPU @ 2.8GHz

Single thread

Self collision is not vectorised: potential ~3x cost savings

Further quality / performance tradeoffs are possible

Issues and Future Work

Performance left on the table

-> SIMD for self collision

Contact Caching is naïve

-> Artist should be able to author contacts?

No viable solution for non-layered clothing

-> True CCD, and GPU

Thank You

Acknowledgements

Arvid Burstrom

Tom Waterson

Paddy O'Brien

Robin Taillandier

Harry Steggles

Frostbite.Team.Physics

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